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Sheaf cohomology

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Grau en Matemàtiques

Introduction

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Part I

Category theory

Chapter 1

Categories, morphisms and duality

1.1 Categories

Definition 1.1.1 (Category)**.** A **category** C consists of:

- a collection $\mathrm{Ob}(\mathsf{C})$ of **objects**.
- for each pair of objects $X, Y \in \mathrm{Ob}(\mathsf{C})$ a collection $\mathrm{Hom}_{\mathsf{C}}(X, Y)$ of **morphisms from** X **to** Y.
- for each object $X \in \mathrm{Ob}(\mathsf{C})$ a morphisms $\mathbb{1}_X \in \mathrm{Hom}_{\mathsf{C}}(X, X)$ called the **identity** morphism.
- for each pair of morphisms $f \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ and $g \in \mathrm{Hom}_{\mathsf{C}}(Y, Z)$ a morphism $g \circ f \in \mathrm{Hom}_{\mathsf{C}}(X, Z)$ called their **composite**.

satisfying the following axioms:

- Composition is associative: for each quadruple $X, Y, Z, W \in \mathrm{Ob}(\mathsf{C})$ of objects, if $f \in \mathrm{Hom}_{\mathsf{C}}(Y, Z)$, $g \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ and $h \in \mathrm{Hom}_{\mathsf{C}}(W, X)$ then
$$(f \circ g) \circ h = f \circ (g \circ h)$$
- Composition satiesfies the unit laws: for each pair of objects $X, Y \in \mathrm{Ob}(\mathsf{C})$ if $f \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ then
$$\mathbb{1}_Y \circ f = f = f \circ \mathbb{1}_X$$

⌟

Notation. To indicate that X is an object of a category C we write $X \in \mathsf{C}$ instead of $X \in \mathrm{Ob}(\mathsf{C})$.

To indicate that $f \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ is a morphisms from X to Y we write $f : X \to Y$ or $X \xrightarrow{f} Y$.

Example 1.1.1.

- The category Set has the collection of all sets as $\mathrm{Ob}(\mathsf{Set})$ and, for any pair of sets X, Y, $\mathrm{Hom}_{\mathsf{Set}}(X, Y)$ is the set of all applications $f : X \to Y$.
- The category Grp has the collection of all groups as $\mathrm{Ob}(\mathsf{Grp})$ and, for any pair of groups G, H $\mathrm{Hom}_{\mathsf{Grp}}(G, H)$ is the set of all group homomorphisms $f : G \to H$.

- The category Top has the collection of all topological spaces as $\mathrm{Ob}(\mathsf{Top})$ and, for any pair of topological spaces X, Y $\mathrm{Hom}_{\mathsf{Top}}(X, Y)$ is the set of all continuous functions $f : X \to Y$.

$\Diamond$

Definition 1.1.2 (Locally small category)**.** A category C is **locally small** if for every pair of objects $X, Y \in \mathsf{C}$ the collection $\mathrm{Hom}(X, Y)$ is a set. ⌟

Definition 1.1.3 (Small category)**.** A locally small category C is **small** if its collection of objects $\mathrm{Ob}(\mathsf{C})$ is a set. ⌟

1.2 Types of morphisms

Definition 1.2.1. ⌟

Definition 1.2.2 (Isomorphism)**.** An **isomorphism** in a category is a morphism $f : X \to Y$ for which exists another morphism $g : Y \to X$ such that:

$$g \circ f = \mathbb{1}_X \quad \text{and} \quad f \circ g = \mathbb{1}_Y$$

The morphism g is called the **inverse** of f, often denoted f^{-1}. ⌟

Example 1.2.1.

- In Set, an isomorphism is a bijection.
- In Grp, an isomorphism is a bijective group homomorphism.
- In Top, an isomorphism is a homeomorphism.

$\Diamond$

Proposition 1.2.1. *Let $f : X \to Y$, $g : Y \to Z$ be two isomorphisms in a category C. Then:*

1. *The inverse f^{-1} of f is unique.*
2. *$g \circ f$ is an isomorphism.*
3. *If $h \circ k$ is an isomorphism and either g or h is an isomorphism then so is the other.*

1.3 Duality

Definition 1.3.1 (Opposite category)**.** Let C be a category. The **opposite category** C^{op} consists of:

- The same objects as C: $\mathrm{Ob}(\mathsf{C}^{\mathrm{op}}) = \mathrm{Ob}(\mathsf{C})$.
- For each pair of objects $X, Y \in \mathsf{C}^{\mathrm{op}}$: $\mathrm{Hom}_{\mathsf{C}^{\mathrm{op}}}(X, Y) = \mathrm{Hom}_{\mathsf{C}}(Y, X)$.

⌟

Example 1.3.1. Consider a preordered set $(P, \leq)$ viewed as a small category where a unique morphism $x \to y$ exists if and only if $x \leq y$.

The opposite category P^{op} corresponds to the same set P equipped with the reversed order $\geq$. That is, a morphism $x \to y$ exists in P^{op} if and only if $y \leq x$ in P. $\Diamond$

Chapter 2

Functors and natural transformations

2.1 Functors

Definition 2.1.1 (Functor)**.** ⌟

2.2 Natural transformations

2.3 Representable functors

Chapter 3

Limits and colimits

3.1 Diagrams, limits and colimits

3.2 Initial and terminal objects

3.3 Products and coproducts

3.4 Directed and inverse limits

Chapter 4

More on functors

4.1 Adjunctions

4.2 Functor categories

Part II

Homological algebra

Chapter 5

Additive and abelian categories

5.1 Additive categories

Definition 5.1.1 (Additive category)**.** A category A is **additive** if it verifies:

A.1) There exists a zero object $0 \in \mathsf{A}$.

A.2) A has finite categorical products.

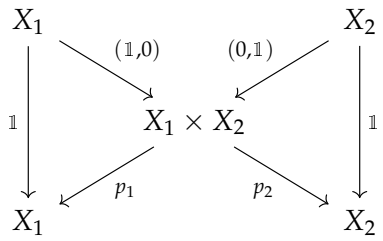
A.3) For every pair of objects $X, Y \in \mathsf{A}$ the set $\mathrm{Hom}_{\mathsf{A}}(X, Y)$ is endowed with an abelian group structure and the composition maps are bilinear with respect to the group operation.

⌟

Lemma 5.1.1. *Finite products and coproducts coincide in an additive category.*

Proof. Consider $X_1, X_2 \in \mathsf{A}$. We only need to check that $X_1 \times X_2 \cong X_1 \amalg X_2$ since the case for a finite collection of elements follows directly by induction. To do this, we will see that $X_1 \times X_2$ satisfies the universal property of the coproduct.

Consider the following diagram



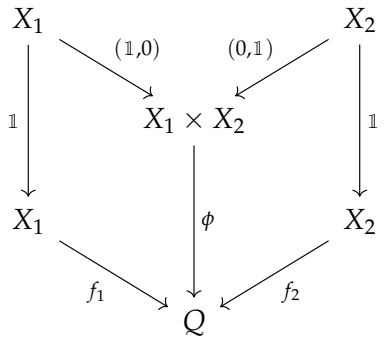
Note that for any pair of morphisms $(\alpha_1, \beta_1), (\alpha_2, \beta_2) : R \to X_1 \times X_2$ we have that

$$(\alpha_1, \beta_1) + (\alpha_2, \beta_2) = (\alpha_1 + \alpha_2, \beta_1, \beta_2)$$

Therefore if p_1, p_2 are two projections on $X_1 \times X_2$ whose sum is the identity we have that

$$(\mathbb{1}, 0) \circ p_1 + (0, \mathbb{1}) \circ p_2 = \mathbb{1}_{X_1 \times X_2}$$

Consider now two morphisms $f_i : X_i \to Q$. Then its sum $\phi = f_1 \circ p_1 + f_2 \circ p_2$ produces the following diagram



Furthermore, this is unique. Indeed, consider $\phi' : X_1 \times X_2 \to Q$ another morphism filling the diagram. Then

$$\begin{aligned}
\phi &= \phi \circ \mathbb{1}_{X_1 \times X_2} \\
&= \phi \circ ((\mathbb{1}, 0) \circ p_1 + (0, \mathbb{1}) \circ p_2) \\
&= \phi \circ (id, 0) \circ p_1 + \phi \circ (0, \mathbb{1}) \circ p_2 \\
&= f_1 \circ p_1 + f_2 \circ p_2 \\
&= \phi' \circ (\mathbb{1}, 0) + \phi' \circ (0, \mathbb{1}) \\
&= \phi' \circ ((\mathbb{1}, 0) \circ p_1 + (0, \mathbb{1}) \circ p_2) \\
&= \phi' \circ \mathbb{1}_{X_1 \times X_2} \\
&= \phi'
\end{aligned}$$

Therefore $X_1 \times X_2$ verifies the universal property of $X_1 \amalg X_2$. □

Notation. If $\{X_i\}_{i=1}^n$ is a finite set of objects in an additive category we will denote its product (and therefore by Lemma 5.1.1 its coproduct) by $\bigoplus_{i=1}^n X_i$ and call it its **direct sum**.

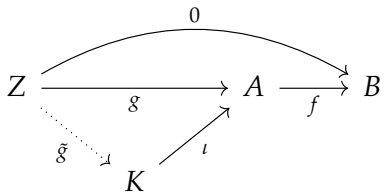
Lemma 5.1.2. *A morphism $\varphi : A \to B$ in an additive category is a monomorphism if and only if for every morphism $\xi : Z \to A$*

$$\varphi \circ \xi = 0 \implies \xi = 0$$

Dually φ is an epimorphism if and only if for every morphism $\xi : B \to Z$

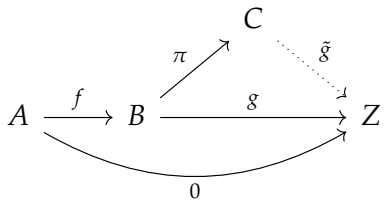
$$\xi \circ \varphi = 0 \implies \xi = 0$$

Definition 5.1.2 (Kernel)**.** Let $f : A \to B$ be a morphism in an additive category. A morphism $\iota : K \to A$ is a **kernel** of f if $f \circ \iota = 0$ and for every morphism $g : Z \to A$ such that $f \circ g = 0$ there exists a unique $\tilde{g} : Z \to K$ making the following diagram commute



⌟

Definition 5.1.3 (Cokernel)**.** Let $f : A \to B$ be a morphism in an additive category. A morphism $\pi : B \to C$ is a **cokernel** of f if $\pi \circ f = 0$ and for every morphism $g : B \to Z$ such that $\pi \circ g = 0$ there exists a unique $\tilde{g} : C \to Z$ making the following diagram commute

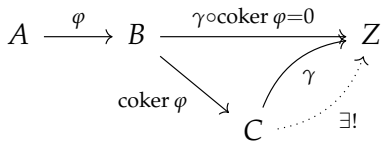


⌟

Lemma 5.1.3. *In an additive category kernels are monomorphisms and cokernels are epimorphisms.*

Proof. We will only show that cokernels are epimorphisms. The statement about kernels follows analogously.

Let $\varphi : A \to B$ be a morphism and let $\operatorname{coker} \varphi : B \to C$ be its cokernel. Consider another morphism $\gamma : C \to Z$ such that $\gamma \circ \operatorname{coker} \varphi = 0$. Since $(\gamma \circ \operatorname{coker} \varphi) \circ \varphi = 0$ by definition of the cokernel we have



Therefore since $\gamma \circ \operatorname{coker} \varphi = 0 = 0 \circ \operatorname{coker} \varphi$ uniqueness forces $\gamma = 0$ and thus by Lemma 5.1.2 φ is an epimorphism. □

Lemma 5.1.4. *Let $\varphi : A \to B$ be a morphism in an additive category. Then φ is a monomorphism if and only if $0 \to A$ is its kernel and φ is an epimorphism if and only if $B \to 0$ is its cokernel.*

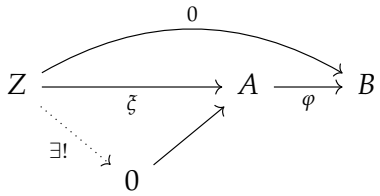
Proof. We will do the kernel one.

$\Longrightarrow$ Suppose φ is a monomorphism.

If $\xi : Z \to A$ is a morphism such that $\varphi \circ \xi = 0$ then by Lemma 5.1.2 $\xi = 0$. In particular ξ factors uniquely through $0 \to A$, i.e: $0 \to A$ is the kernel of φ.

$\Longleftarrow$ Suppose now that $0 \to A$ is the kernel of φ.

Consider a morphism $\xi : Z \to A$ such that $\varphi \circ \xi = 0$. By the kernel properties we have the following diagram



Therefore by uniqueness $\xi = 0$, making φ a monomorphism by Lemma 5.1.2.

□

5.1.1 Additive functors

Definition 5.1.4 (Additive functor)**.** Let A, B be two additive categories. A functor $F : \mathsf{A} \to \mathsf{B}$ is **additive** if it preserves the abelian group structure of all Hom-sets. ⌟

5.2 Abelian categories

Definition 5.2.1 (Abelian category)**.** An additive category A is **abelian** if it verifies

A.4) Kernels and cokernels exist in A.

A.5) Every monomorphism is the kernel of some morphism and every epimorphism is the cokernel of some morphism.

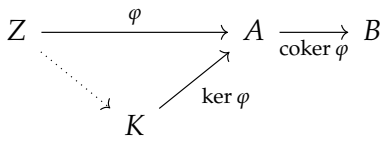
⌟

Lemma 5.2.1. *In an abelian category* A *every kernel is the kernel of its cokernel and every cokernel is the cokernel of its kernel. In formulas*

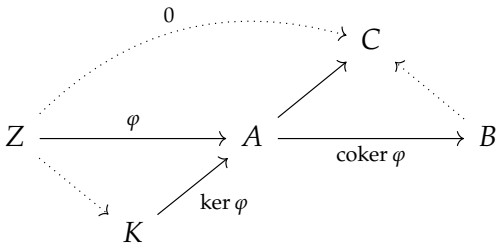
$$\ker(\operatorname{coker}\varphi) = \ker\varphi \qquad \operatorname{coker}(\ker\varphi) = \operatorname{coker}\varphi$$

Proof. Lets check cokernels this time.

Let $\operatorname{coker}\varphi : A \to B$ be the cokernel of a morphism $\varphi : Z \to A$. Since A is abelian φ has a kernel $\ker\varphi : K \to A$. Note that $Z \to A \to B$ is 0, therefore by the kernel universal property we have



Consider a morphism $A \to C$ such that $(K \to A \to C) = 0$. Now using the cokernel universal property we obtain

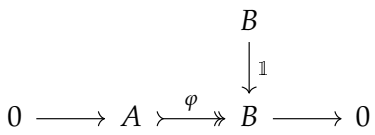


Therefore $\operatorname{coker}\varphi$ satisfies the universal property of the cokernel of $\ker\varphi : K \to A$. □

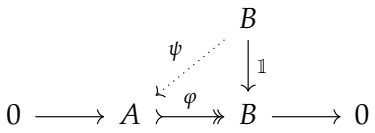
Lemma 5.2.2. *Let $\varphi : A \to B$ be a morphism in an abelian category* A*. If φ is both a monomorphism and epimorphism then φ is an isomorphism.*

Proof. By Lemma 5.1.4 the kernel of φ is $0 \to A$ and its cokernel is $B \to 0$. Furthermore by Lemma 5.2.1 φ is the cokernel of $0 \to A$ and the kernel of $B \to 0$.

Consider the following diagram

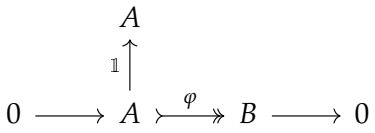


Since $(B \to B \to 0) = 0$ and φ is the kernel of $B \to 0$ we have

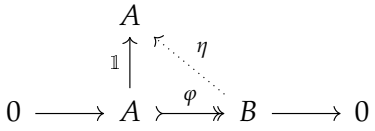


Therefore since $\varphi\psi = \mathbb{1}$ we have shown that φ has a right-inverse.

Consider now the diagram



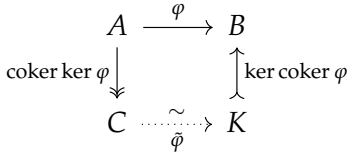
Since $(0 \to A \to A) = 0$ and φ is the cokernel of $0 \to A$ we have



Therefore since $\eta\varphi = \mathbb{1}$ we have shown that φ has a left-inverse. □

The factorization property

Theorem 5.2.1 (Factorization property)**.** *Every morphism $\varphi : A \to B$ in an abelian category* A *can be decomposed as*



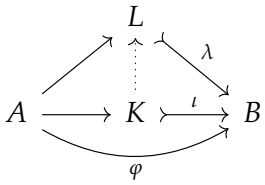
Before proving this statement, we need to consider some auxiliary lemmas.

Lemma 5.2.3. *Let $\varphi : A \to B$ be a morphism in an abelian category and let $\iota : K \to B$ be the kernel of its cokernel. Then*

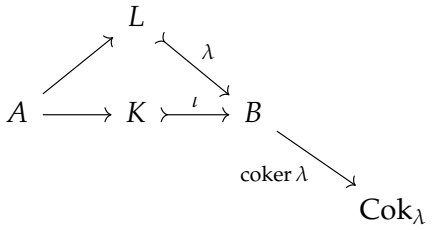
- *ι is a monomorphism.*
- *φ factors through ι.*
- *ι is initial with these properties.*

Proof.

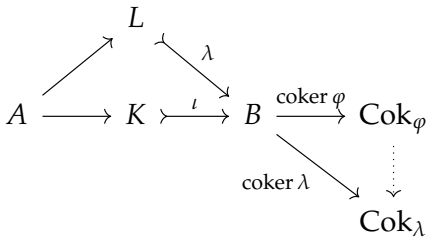
- By Lemma 5.1.3 ι is a monomorphism.
- Since $A \to B \to \operatorname{coker}\varphi$ is the zero morphism by the universal property of kernels there is an induced morphism $K \to A$.
- We want to see that if $\lambda : L \rightarrowtail B$ is a monomorphism that factors through φ then ι factors uniquely through λ



Consider coker λ



Since φ factors through λ we have that $A \to B \to \mathrm{Cok}_\lambda = 0$. Therefore by the universal property of the cokernel we have an induced map



Therefore since $K \to \mathrm{Cok}_\varphi = 0$ we have that $K \to B \xrightarrow{\operatorname{coker}\lambda} \mathrm{Cok}_\lambda = 0$.

Finally since by Lemma 5.2.1 $\lambda : L \to B$ is the kernel of its cokernel there exists a unique morphism $K \to L$ making the diagram commute.

□

Definition 5.2.2. Let $\varphi : A \to B$ be a morphism in an abelian category. The **image** of φ is

$$\operatorname{im}\varphi = \ker\operatorname{coker}\varphi$$

and the **coimage** of φ is

$$\operatorname{coim}\varphi = \operatorname{coker}\ker\varphi$$

⌟

Notation. If $\varphi : A \rightarrowtail B$ is a monomorphism we denote by ${}^{B}\!/_{A}$ the image of coker φ.

In this way the target of coim φ is ${}^{B}\!/_{\ker\varphi}$.

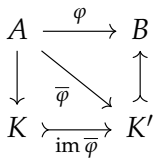
Lemma 5.2.4. *Let $\varphi : A \to B$ be a morphism in an abelian category with $\operatorname{im}\varphi : K \to B$ and $\operatorname{coim}\varphi : A \to C$. Then the induced morphisms $A \to K$ and $B \to C$ are, respectively, an epimorphism and a monomorphism.*

Proof. We will see that $\overline{\varphi} : A \to K$ is a epimorphism. The other statement follows analogously.

Consider the image of $\overline{\varphi}$

$$K' \xrightarrow{\operatorname{im}\overline{\varphi}} K \xrightarrow{\operatorname{coker}\overline{\varphi}} \mathrm{Cok}$$

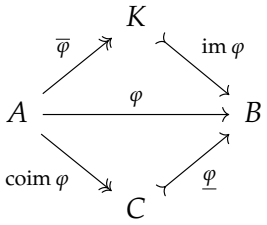
Since by Lemma 5.2.1 every epimorphism is the cokernel of its kernel we have that $\operatorname{coker} \overline{\varphi} = \operatorname{coker} \operatorname{im} \overline{\varphi}$. Moreover since $K' \rightarrowtail K$ is a monomorphism through which $\overline{\varphi}$ factors we have



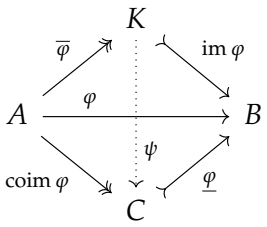
Since $\operatorname{im} \varphi$ is initial among this kind of morphisms by Lemma 5.2.3 necessarily $\operatorname{im} \overline{\varphi}$ is an isomorphism.

Therefore $0 = \operatorname{coker} \operatorname{im} \overline{\varphi} = \operatorname{coker} \overline{\varphi}$, proving that $\overline{\varphi}$ is an epimorphism by Lemma 5.1.4. □

To summarize, we have the following situation



Proof of Theorem 5.2.1. By the universal property of $\operatorname{im} \varphi$ and $\operatorname{coim} \varphi$ there exists a unique morphism $\psi : K \to C$ making the above diagram commute



Since $\varphi \circ \psi = \operatorname{im} \varphi$ is a monomorphism so is ψ. Since $\psi \circ \overline{\varphi} = \operatorname{coim} \varphi$ is a epimorphism so is ψ. Therefore by Lemma 5.2.2 ψ is an isomorphism.

Taking $\tilde{\varphi} = \psi^{-1} : C \to K$ we are done. □

5.3 Exactness

Definition 5.3.1 (Exact sequence)**.** An sequence of morphisms

$$X \xrightarrow{f} Y \xrightarrow{g} Z$$

in an abelian category A is **exact** at Y if $f \circ g = 0$ and $\operatorname{im} f = \ker g$. ⌟

Definition 5.3.2 (Short exact sequence)**.** An sequence of morphisms

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$

in an abelian-category A is **exact** is called a **short exact sequence**.

Exact complexes of the form $0 \to X \to Y \to Z$ (resp. $X \to Y \to Z \to 0$) are called left (resp. right) short exact sequences. ⌟

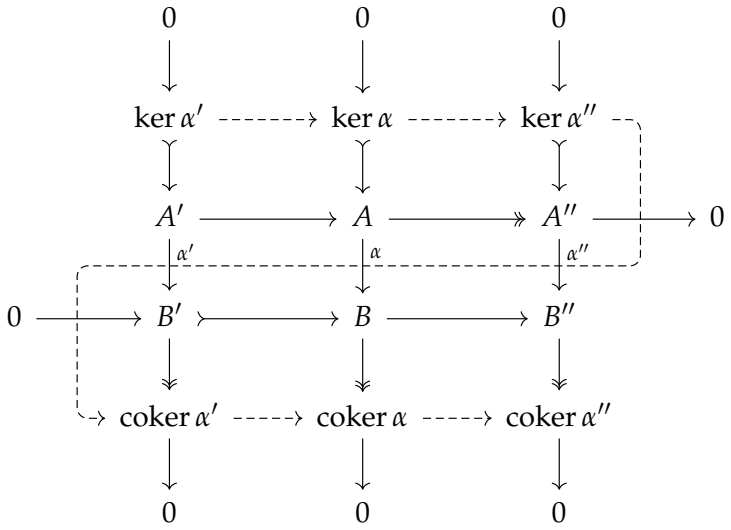
Remark 5.3.1. For any morphism $f : X \to Y$ in an A we get short exact sequences

$$0 \longrightarrow \ker f \longrightarrow X \longrightarrow \operatorname{im} f \longrightarrow 0$$

$$0 \longrightarrow \operatorname{im} f \longrightarrow Y \longrightarrow \operatorname{coker} f \longrightarrow 0$$

Therefore by Lemma 5.1.4 the sequence $0 \to X \xrightarrow{f} Y$ (resp. $X \xrightarrow{f} Y \to 0$) is exact if and only if f is a monomorphism (resp. epimorphism). ▽

Theorem 5.3.1 (Snake lemma)**.** *In any abelian category, any diagram of the form*

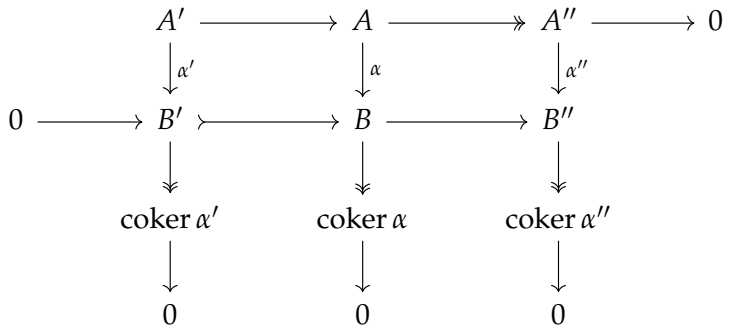


with exact rows and columns can be completed by dashed arrows making this sequence exact.

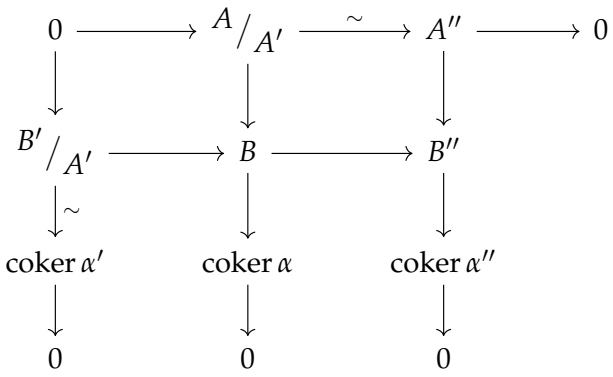
Proof. We will give a purely *arrow-theoretic* proof following [Wis].

We will be using Lemma 5.1.3 and Lemma 5.2.2 throughout the proof without explicit mention.

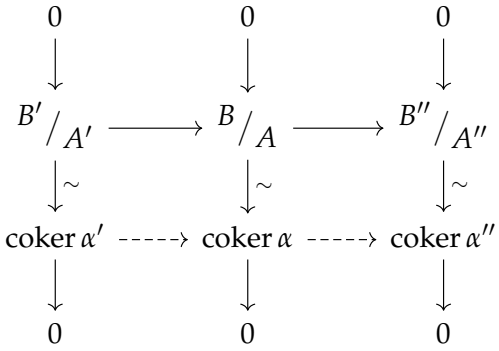
Consider first the diagram



Take first cokernels by A'



And then take cokernels by A/A'



This produces an exact sequence $\operatorname{coker}\alpha' \to \operatorname{coker}\alpha \to \operatorname{coker}\alpha''$.

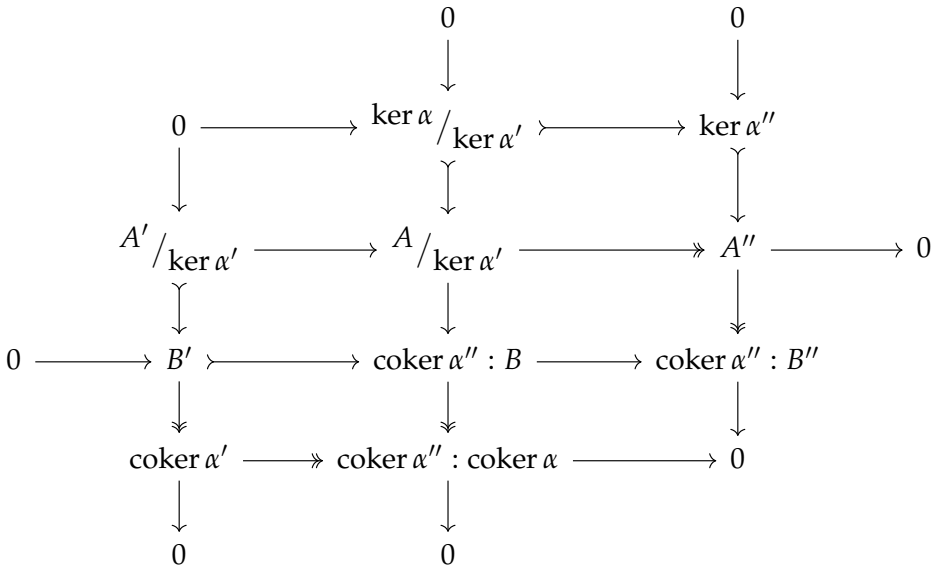
Taking kernels instead we get an exact sequence $\ker\alpha' \to \ker\alpha \to \ker\alpha''$.

It remains to produce a map $\delta : \ker\alpha'' \to \operatorname{coker}\alpha'$ and show that it induces an isomorphism

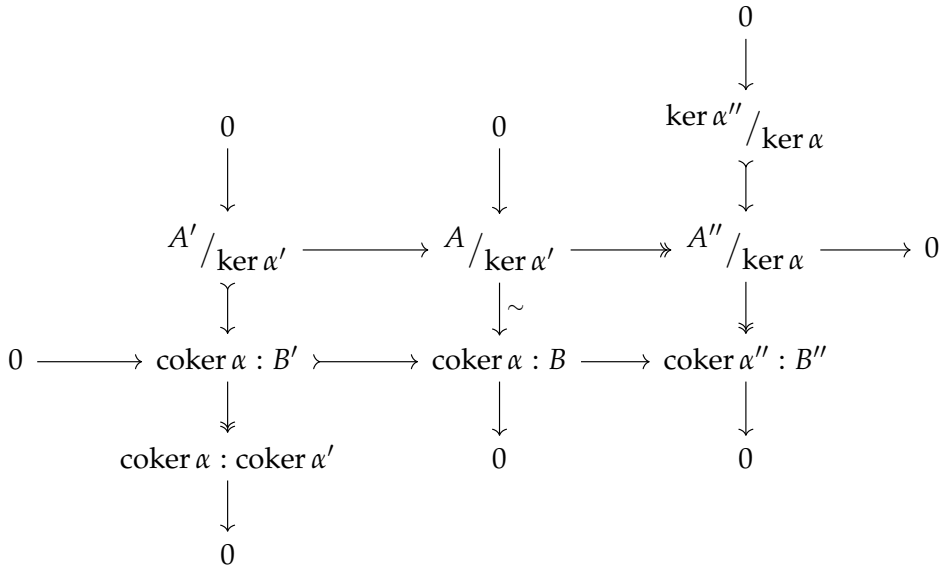
$$\operatorname{coker}(\ker\alpha \to \ker\alpha'') \cong \ker(\operatorname{coker}\alpha \to \operatorname{coker}\alpha')$$

To simplify the notation, we will denote $\ker(\operatorname{coker}\alpha \to \operatorname{coker}\alpha'')$ by $\operatorname{coker}\alpha : \operatorname{coker}\alpha''$.

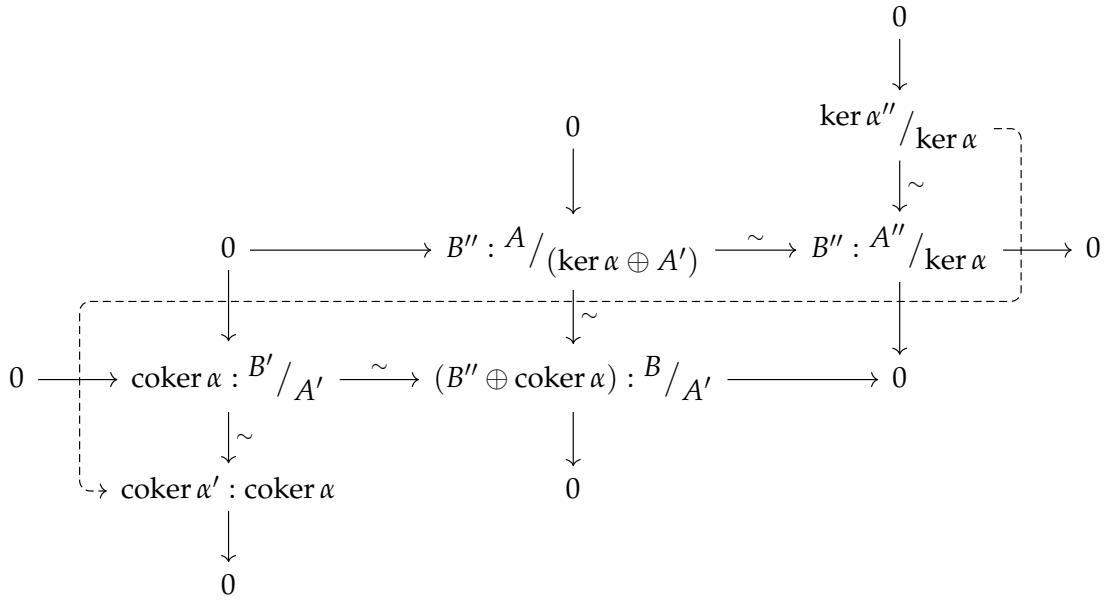
First, take cokernels by $\ker\alpha'$ and kernels into $\operatorname{coker}\alpha''$ in the full diagram



Now, take cokernels by $\ker\alpha/\ker\alpha'$ and kernels into $\operatorname{coker}\alpha : \operatorname{coker}\alpha''$



Finally, taking cokernels by $A'/\ker\alpha'$ and kernels into $\operatorname{coker}\alpha'' : B''$ we get



□

5.3.1 Exact functors

Definition 5.3.3 (Exact functor)**.** Let A, B be two abelian categories.

An additive functor $F : \mathsf{A} \to \mathsf{B}$ is called left (resp. right) **exact** if for any short exact sequence in A

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z$$

(resp. $X \to Y \to Z \to 0$) the squence

$$0 \longrightarrow F(X) \longrightarrow F(Y) \longrightarrow F(Z)$$

(resp. $F(X) \to F(Y) \to F(Z) \to 0$) is exact.

If F is both left and right exact, then F is called **exact**. ⌟

Example 5.3.1. The functors $\mathrm{Hom}_{\mathsf{A}}(X, -)$ and $\mathrm{Hom}_{\mathsf{A}}(-, X)$ are both left-exact functors from A to Ab. ◊

Proof. We will check the statement about $\mathrm{Hom}_{\mathsf{A}}(X, -)$, the other follows analogously.

Consider a short left-exact sequence

$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C$$

in A. We want to show that

$$0 \longrightarrow \mathrm{Hom}(X, A) \xrightarrow{f_*} \mathrm{Hom}(X, B) \xrightarrow{g_*} \mathrm{Hom}(X, C)$$

is exact in Ab.

By Remark 5.3.1, to show exacteness at $\mathrm{Hom}(X, A)$ we need to check that f_* is a monomorphism in Ab, i.e: f_* is injective.

Let $\alpha \in \mathrm{Hom}(X, A)$ such that $f_*(\alpha) = f \circ \alpha = 0$. Since $0 \to A \xrightarrow{f} B$ is exact by **??** f is a monomorphism. Therefore $f \circ \alpha = 0 \iff \alpha = 0$, proving that f_* is injective.

To see exactness at $\mathrm{Hom}(X, B)$ we are going to check that $\mathrm{im}\, f_* = \ker g_*$:

- $\subseteq$ Let $\phi = f_*(\alpha)$ for some $\alpha \in \mathrm{Hom}(X, A)$. Applying g_* we get $g_*(\phi) = (g \circ f) \circ \alpha$, therefore by exactness of the original sequence $\phi \in \ker g_*$.
- $\supseteq$ Let $\beta \in \mathrm{Hom}(X, B)$ such that $g_*(\beta) = 0$. Since the original sequence is exact $\ker g = f$, therefore by the universal property of the kernel there exists $\alpha : X \to A$ such that $f \circ \alpha = \beta$.

□

The following is [Sch, Lemma 5.2.2].

Lemma 5.3.1. *Let $F : \mathsf{A} \to \mathsf{B}$ be an additive functor.*

1. *The following are equivalent*
   (a) *F is left exact.*
   (b) *F commutes with kernels: for any morphism $f : X \to Y$, $F(\ker f) \xrightarrow{\sim} \ker F(f)$.*
   (c) *F commutes with finite limits.*
   (d) *For any short exact sequence $0 \to X \to Y \to Z \to 0$ in A the sequence $0 \to F(X) \to F(Y) \to F(Z)$ is exact in B.*
2. *The following are equivalent*
   (a) *F is exact.*
   (b) *For any exact sequence $X \to Y \to Z$ in A the sequence $F(X) \to F(Y) \to F(Z)$ is exact in B.*
   (c) *F commutes with finite limits and finite colimits.*

Chapter 6

Complexes and cohomology

6.1 Complexes and cohomology

Throughout this chapter we will be working over a fixed abelian category A.

Definition 6.1.1 (Complex)**.** A sequence of objects and morphisms

$$\cdots \xrightarrow[d_{i+2}]{} C_{i+1} \xrightarrow[d_{i+1}]{} C_i \xrightarrow[d_i]{} C_{i-1} \xrightarrow[d_{i-1}]{} \cdots$$

in A is a **chain complex** if for every $i \in \mathbb{Z} \quad d_i \circ d_{i+1} = 0$. We denote a chain complex as $(C_\bullet, d_\bullet)$ or simply as $C_\bullet$.

By reversing the order, we say that a sequence

$$\cdots \xrightarrow{d^{i-2}} C^{i-1} \xrightarrow{d^{i-1}} C^i \xrightarrow{d^i} C^{i+1} \xrightarrow{d^{i+1}} \cdots$$

is a **cochain complex** if for every $i \in \mathbb{Z} \quad d^i \circ d^{i-1} = 0$. We denote a cochain complex as $(C^\bullet, d^\bullet)$ or simply as $C^\bullet$. ⌟

Remark 6.1.1. In purelly algebraic terms, chains are cochains are *equivalent*. We will adopt the cochain notation, but all proofs and definitions translate to the chain notation. Therefore from now onwards we will be refering to *cochains* as *chains* or *sequences*. ▽

Definition 6.1.2 (Morphism of complexes)**.** Let $C^\bullet, D^\bullet$ be two complexes in A. A **morphism of complexes** $f : C^\bullet \to D^\bullet$ is a collection of morphisms $\{f^i : C^i \to D^i\}_{i\in\mathbb{Z}}$ making, for every $i \in \mathbb{Z}$, the following diagram commute

$$\begin{array}{ccc} C^i & \xrightarrow{d^i_C} & C^{i+1} \\ {\scriptstyle f^i}\downarrow & & \downarrow{\scriptstyle f^{i+1}} \\ D^i & \xrightarrow{d^i_D} & D^{i+1} \end{array}$$

⌟

Definition 6.1.3. Given any abelian category A we can construct a **complex category** $\mathsf{Ch}(\mathsf{A})$ by taking $\mathrm{Ob}_{\mathsf{Ch}(\mathsf{A})}$ to be the collection of all complexes with values in A and, for every $C^\bullet, D^\bullet \in \mathrm{Ob}_{\mathsf{Ch}(\mathsf{A})}$, $\mathrm{Hom}_{\mathsf{Ch}(\mathsf{A})}(C^\bullet, D^\bullet)$ to be the collection of all morphisms of complexes between $C^\bullet$ and $D^\bullet$. ⌟

Remark 6.1.2. A morphism of complexes $f : C^\bullet \to D^\bullet$ is a monomorphism (resp. epimorphism) if for every $i \in \mathbb{Z}$ the morphism $f^i : C^i \to D^i$ is a monomorphism (resp. epimorphism) in A. ▽

Theorem 6.1.1. *If* A *is an abelian-category, so is its category of complexes* $\mathsf{Ch}(\mathsf{A})$.

Proof. By taking addition *elementwise* and $0^\bullet$ as the zero object we can make $\mathsf{Ch}(\mathsf{A})$ into an additive category. Therefore we only need to prove that $\mathsf{Ch}(\mathsf{A})$ verifies the axioms of an abelian category.

Let $f : A^\bullet \to B^\bullet$ be a morphism of complexes.

Since A is abelian, for every $i \in \mathbb{Z}$ the map $f^i : A^i \to B^i$ admits a kernel $\ker f^i : K^i \rightarrowtail A^i$ and a cokernel $\operatorname{coker} f^i : B^i \twoheadrightarrow C^i$.

Consider the diagram

$$\begin{array}{ccc}
K^i & & K^{i+1} \\
{\scriptstyle \ker f^i}\downarrow & & \downarrow{\scriptstyle \ker f^{i+1}} \\
A^i & \xrightarrow{d_C^i} & A^{i+1} \\
{\scriptstyle f^i}\downarrow & & \downarrow{\scriptstyle f^{i+1}} \\
B^i & \xrightarrow{d_D^i} & B^{i+1} \\
{\scriptstyle \operatorname{coker} f^i}\downarrow & & \downarrow{\scriptstyle \operatorname{coker} f^{i+1}} \\
C^i & & C^{i+1}
\end{array}$$

Since the inner square is commutative we get unique morphisms $K^i \to K^{i+1}$ and $C^i \to C^{i+1}$ making the whole diagram commute, obtaining chain complexes $K^\bullet$ and $C^\bullet$ and morphisms of chains $\ker f^\bullet, \operatorname{coker} f^\bullet$. It is then easy to check that $\ker f^\bullet$ and $\operatorname{coker} f^\bullet$ verify the corresponding universal properties.

On the other hand, by Remark 6.1.2 the other axiom is easily verified *element by element*. □

Definition 6.1.4 (Bounded complexes)**.** Let $C^\bullet$ be a complex. We say that $C^\bullet$ is

- **bounded below** if there exists $k \in \mathbb{Z}$ such that $\forall i \leq k \quad C^i = 0$.
- **bounded above** if there exists $k \in \mathbb{Z}$ such that $\forall i \geq k \quad C^i = 0$.
- **bounded** if it is both bounded below and above.

⌟

Notation. We denote by $\mathsf{Ch}^{-,+,b}(\mathsf{A}) \hookrightarrow \mathsf{Ch}(\mathsf{A})$ the full subcategory of the chain complex category consisting of bounded below, bounded above and bounded complexes respectively.

We also have variants $\mathsf{Ch}^{\geq 0,\leq 0}(\mathsf{A}) \hookrightarrow \mathsf{Ch}(\mathsf{A})$ of complexes with $\forall i \geq 0$ (resp $\forall i \leq 0$) $C^i = 0$.

By Theorem 6.1.1 all of this categories are abelian.

Remark 6.1.3. For all integers $i \in \mathbb{Z}$ there exists a fully faithfull functor

$$\iota_r : \mathsf{A} \to \mathsf{Ch}(\mathsf{A})$$

sending an object $A \in \mathsf{A}$ to a complex

$$\cdots \longrightarrow 0 \xrightarrow{d^{r-1}} A \xrightarrow{d^r} 0 \longrightarrow \cdots$$

and a morphism $\varphi : A \to B$ in A to a morphism of complexes

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & 0 & \xrightarrow{d^{r-1}} & A & \xrightarrow{d^r} & 0 & \longrightarrow & \cdots \\ & & \downarrow & & \downarrow{\scriptstyle \varphi} & & \downarrow & & \\ \cdots & \longrightarrow & 0 & \xrightarrow{d^{r-1}} & B & \xrightarrow{d^r} & 0 & \longrightarrow & \cdots \end{array}$$

In this way, we can identify A as its *image* via $\iota = \iota_0$ in $\mathsf{Ch}(\mathsf{A})$. ▽

Definition 6.1.5 (Shift functor)**.** For any integer $i \in \mathbb{Z}$ we define the **shift** functor

$$\begin{aligned} \mathsf{Ch}(\mathsf{A}) &\to \mathsf{Ch}(\mathsf{A}) \\ M^\bullet &\mapsto M[i]^\bullet \end{aligned}$$

by $M[i]^k = M^{k+i}$ and $d^k_{M[i]^\bullet} = (-1)^i d^{k+i}_{M^\bullet}$. ⌟

Definition 6.1.6 (Exact complex)**.** A complex $C^\bullet$ is **exact** at position i if the sequence

$$C^{i-1} \xrightarrow{d^{i-1}} C^i \xrightarrow{d^i} C^{i+1}$$

is exact.

If the complex is exact at every position, we say that it is **exact**. ⌟

6.2 Cohomology

Definition 6.2.1 (Cohomology)**.** The ***i*-th cohomology** of a complex $C^\bullet$ is

$$\mathrm{H}^i(C^\bullet) = {\ker d^i}\big/{\operatorname{im} d^{i-1}}$$

⌟

Remark 6.2.1. The definition of cohomology should be parsed as follows: let $I \to C^i$ be $\operatorname{im} d^{i-1}$ and $K \to C^i$ be $\ker d^i$. Since $C^\bullet$ is a complex $\operatorname{im} d^{i-1}$ factors through $\ker d^i$, giving a monomorphism $I \rightarrowtail K$. Then $\mathrm{H}^i(C^\bullet) = \operatorname{coker}(I \rightarrowtail K)$. ▽

Proposition 6.2.1. *For every integer i, the assignement $\mathrm{H}^i : M^\bullet \mapsto H^i(M^\bullet)$ taking a complex $M^\bullet$ to its i-th cohomology group is a functor $\mathsf{Ch}(\mathsf{A}) \to \mathsf{A}$.*

Proof. We want to show that for all complexes $M^\bullet, N^\bullet \in \mathsf{Ch}(\mathsf{A})$ and $i \in \mathbb{Z}$, H^i induces in a natural way a homomorphism of abelian groups

$$\operatorname{Hom}_{\mathsf{Ch}(\mathsf{A})}(M^\bullet, N^\bullet) \to \operatorname{Hom}_{\mathsf{A}}(\mathrm{H}^i(M^\bullet), \mathrm{H}^i(N^\bullet))$$

Let $\alpha^\bullet : M^\bullet \to N^\bullet$ be a morphism of complexes. Let's consider the relevant part of the commutative diagram defining $\alpha^\bullet$:

$$\begin{array}{ccccc} M^{i-1} & \xrightarrow{d^{i-1}_{M^\bullet}} & M^i & \xrightarrow{d^i_{M^\bullet}} & M^{i+1} \\ {\scriptstyle \alpha^{i-1}}\downarrow & & {\scriptstyle \alpha^i}\downarrow & & {\scriptstyle \alpha^{i+1}}\downarrow \\ N^{i-1} & \xrightarrow{d^{i-1}_{N^\bullet}} & N^i & \xrightarrow{d^i_{N^\bullet}} & N^{i+1} \end{array}$$

By the commutativity of the right square we have that $d^i_{N^\bullet} \circ \alpha^i = \alpha^{i+1} \circ d^i_{M^\bullet}$ is 0 on $\ker d^i$. Therefore the restriction of α^i to $\ker d^i_{M^\bullet}$ factors through $\ker d^i_{N^\bullet}$. Composing with the projection we get a morphism

$$\ker d^i_{M^\bullet} \to \mathrm{H}^i(N^\bullet)$$

Since this morphism is 0 when restricted to $\operatorname{im} d_{M^\bullet}^{i-1}$ by the commutativity of the left square we get that α^i induces a morphism

$$\mathrm{H}^i(M^\bullet) \to \mathrm{H}^i(N^\bullet)$$

in A as we wanted. □

Remark 6.2.2. By Remark 6.1.3 we can see cohomology as a functor $\mathrm{H}^i : \mathsf{Ch}(\mathsf{A}) \to \mathsf{Ch}(\mathsf{A})$ using ι_i. Note that for this particular complex $M^\bullet$ we have that

$$\mathrm{H}^\bullet(\mathrm{H}^\bullet(M^\bullet)) = \mathrm{H}^\bullet(M^\bullet)$$

▽

Theorem 6.2.1 (Long Exact Sequence in Cohomology)**.** *Every short exact sequence*

$$0 \longrightarrow A^\bullet \xrightarrow{f^\bullet} B^\bullet \xrightarrow{g^\bullet} C^\bullet \longrightarrow 0$$

in $\mathsf{Ch}(\mathsf{A})$ *induces a long exact sequence in cohomology*

$$\begin{array}{l} \cdots \longrightarrow \mathrm{H}^{i-1}(B) \longrightarrow \mathrm{H}^{i-1}(C) \\ \quad \hookrightarrow \mathrm{H}^i(A) \longrightarrow \mathrm{H}^i(B) \longrightarrow \mathrm{H}^i(C) \\ \quad\quad \hookrightarrow \mathrm{H}^{i+1}(A) \longrightarrow \mathrm{H}^{i+1}(B) \longrightarrow \cdots \end{array}$$

Proof. For every integer i, applying the snake lemma to the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A^i & \xrightarrow{f^i} & B^i & \xrightarrow{g^i} & C^i & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle d_A^i} & & \downarrow{\scriptstyle d_B^i} & & \downarrow{\scriptstyle d_C^i} & & \\ 0 & \longrightarrow & A^{i+1} & \xrightarrow{f^{i+1}} & B^{i+1} & \xrightarrow{g^{i+1}} & C^{i+1} & \longrightarrow & 0 \end{array}$$

we get an exact sequence

$$\operatorname{coker} d_A^i \longrightarrow \operatorname{coker} d_B^i \longrightarrow \operatorname{coker} d_C^i \longrightarrow 0$$

Applying now the snake lemma to the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A^{i+1} & \xrightarrow{f^{i+1}} & B^{i+1} & \xrightarrow{g^{i+1}} & C^{i+1} & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle d_A^{i+1}} & & \downarrow{\scriptstyle d_B^{i+1}} & & \downarrow{\scriptstyle d_C^{i+1}} & & \\ 0 & \longrightarrow & A^{i+2} & \xrightarrow{f^{i+2}} & B^{i+2} & \xrightarrow{g^{i+2}} & C^{i+2} & \longrightarrow & 0 \end{array}$$

we get an exact sequence

$$0 \longrightarrow \ker d_A^{i+1} \longrightarrow \ker d_B^{i+1} \longrightarrow \ker d_C^{i+1}$$

Since $d^{i+1} \circ d^i = 0$ we have the following diagram

$$\begin{array}{ccccccc} & & \operatorname{coker} d_A^i & \longrightarrow & \operatorname{coker} d_B^i & \longrightarrow & \operatorname{coker} d_C^i & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \overline{d}_A^i} & & \downarrow{\scriptstyle \overline{d}_B^i} & & \downarrow{\scriptstyle \overline{d}_C^i} & & \\ 0 & \longrightarrow & \ker d_A^{i+1} & \longrightarrow & \ker d_B^{i+1} & \longrightarrow & \ker d_C^{i+1} & & \end{array}$$

Applying the snake lemma to this diagram we get an exact sequence

$$\begin{array}{ccccc} \ker \overline{d}_A^i & \longrightarrow & \ker \overline{d}_B^i & \longrightarrow & \ker \overline{d}_A^i \\ & & & \overset{\delta^i}{\searrow} & \\ \operatorname{coker} \overline{d}_A^i & \longrightarrow & \operatorname{coker} \overline{d}_B^i & \longrightarrow & \operatorname{coker} \overline{d}_C^i \end{array}$$

It only remains to prove that $\ker \overline{d}_A^i \cong \mathrm{H}^i(A)$ and $\operatorname{coker} \overline{d}_A^i \cong \mathrm{H}^{i+1}(A)$.

Consider the diagram

$$\begin{array}{ccccccccc} & & 0 & & 0 & & 0 & & \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \operatorname{im} d_A^{i-1} & = & \operatorname{im} d_A^{i-1} & \longrightarrow & 0 & & \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \ker d_A^i & \longrightarrow & A^i & \longrightarrow & \operatorname{im} d_A^i & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathrm{H}^i(A) & \longrightarrow & {}^{A^i}\!/_{\operatorname{im} d_A^{i-1}} & \xrightarrow{\psi} & \operatorname{im} d_A^i & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ & & 0 & & 0 & & 0 & & \end{array}$$

Since the first two rows are exact by the nine lemma we have a short exact sequence

$$0 \longrightarrow \mathrm{H}^i(A) \longrightarrow {}^{A^i}\!/_{\operatorname{im} d_A^{i-1}} \xrightarrow{\psi} \operatorname{im} d_A^i \longrightarrow 0$$

Therefore since $\overline{d}_A^i : {}^{A^i}\!/_{\operatorname{im} d_A^{i-1}} \xrightarrow{\psi} \operatorname{im} d_A^i \xrightarrow{\iota} \ker d_A^{i+1}$ we have

$$\ker(\overline{d}_A^i) = \ker \psi \cong \mathrm{H}^i(A^\bullet)$$

On the other hand, note that $\operatorname{coker} \overline{d}_A^i = \operatorname{coker}(\iota \circ \psi) = \operatorname{coker} \iota$ and

$$\operatorname{coker} \iota = {}^{\ker d_A^{i+1}}\!/_{\operatorname{im} \iota} = {}^{\ker d_A^{i+1}}\!/_{\operatorname{im} d_A^i} \cong \mathrm{H}^{i+1}(A)$$

□

Triangles

Consider a short exact sequence of complexes

$$0 \longrightarrow A^\bullet \longrightarrow B^\bullet \longrightarrow C^\bullet \longrightarrow 0$$

Note that we can map $C^\bullet$ to $A^\bullet$ by sending all the complex $C^\bullet$ to 0 *within* $A^\bullet$

$$C^\bullet \xrightarrow{0} A^\bullet$$

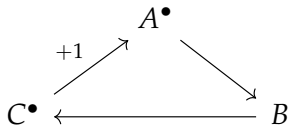
In this way, the exactness of the original sequence is equivalent to the exactness at the center of

$$C^\bullet \xrightarrow{0} A^\bullet \longrightarrow B^\bullet$$

$$A^\bullet \longrightarrow B^\bullet \longrightarrow C^\bullet$$

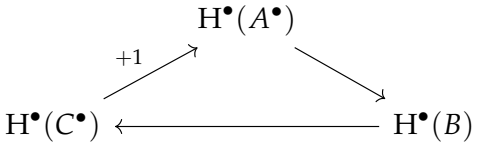
$$B^\bullet \longrightarrow C^\bullet \xrightarrow{0} A^\bullet$$

We can represent this using the *exact triangle*



The arrow marked by $+1$ in this case refers to the zero morphism. We use $+1$ to record that we can view this morphism from $C^\bullet$ to the shift $A[1]^\bullet$. Therefore, the triangle is shorthand for the diagram

Using Theorem 6.2.1 we get another *exact triangle*



However this time the $+1$ morphism is, generally, non-zero.

Chapter 7

Cones and homotopy

7.1 The mapping cone of a morphism

Definition 7.1.1 (Mapping cone)**.** Let $\alpha^\bullet : L^\bullet \to M^\bullet$ be a morphism of complexes. We define the **mapping cone** $\mathrm{Cone}(\alpha)^\bullet$ as the complex with objects

$$\forall i \in \mathbb{Z} \quad \mathrm{Cone}(\alpha)^i = L[1]^i \oplus M^i = L^{i+1} \oplus M^i$$

and differentials

$$\forall i \in \mathbb{Z} \quad d^i_{\mathrm{Cone}(\alpha)^\bullet}(\ell, m) = (-d^{i+1}_{L^\bullet}(\ell), \alpha^{i+1}(\ell) + d^i_{M^\bullet}(m))$$

Pictorially

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & L^{i+1} & \xrightarrow{-d^{i+1}_L} & L^{i+1} & \longrightarrow & \cdots \\ & \searrow & \oplus & \searrow^{\alpha^{i+1}} & \oplus & \searrow & \\ \cdots & \longrightarrow & M^i & \xrightarrow[d^i_M]{} & M^{i+1} & \longrightarrow & \cdots \end{array}$$

⌟

Remark 7.1.1. We have morphisms of complexes $M^\bullet \to \mathrm{Cone}(\alpha)^\bullet$ and $\mathrm{Cone}(\alpha)^\bullet \to L[1]^\bullet$ induced by the morphisms $M^i \to L[1]^i \oplus M^i \to L[1]^i$.

Furthermore, since the we have short exact sequences

$$0 \longrightarrow M^i \longrightarrow L[1]^i \oplus M^i \longrightarrow L[1]^i \longrightarrow 0$$

the sequence of complexes

$$0 \longrightarrow M^\bullet \longrightarrow L[1]^\bullet \oplus M^\bullet \longrightarrow L[1]^\bullet \longrightarrow 0$$

is exact. ▽

Proposition 7.1.1. *There exists an exact triangle*

$$\begin{array}{ccc} & \mathrm{H}^\bullet(M^\bullet) & \\ {\scriptstyle +1}\nearrow & & \searrow \\ \mathrm{H}^\bullet(L[1]^\bullet) & \xleftarrow{\quad\delta\quad} & \mathrm{H}^\bullet(\mathrm{Cone}(\alpha)^\bullet) \end{array}$$

where the connecting morphism δ is the morphism induced by $\alpha^\bullet$ in cohomology.

Proof. The existence of the triangle comes directly from Theorem 6.2.1, so we only need to check that δ agrees with the morphism induced by $\alpha^\bullet$.

By chasing the diagram

$$\begin{array}{ccccccccc}
 & & \vdots & & \vdots & & \vdots & & \\
 & & \uparrow & & \uparrow & & \uparrow & & \\
0 & \longrightarrow & M^i & \longrightarrow & L^{i+1} \oplus M^i & \longrightarrow & L^{i+1} & \longrightarrow & 0 \\
 & & \uparrow & & \uparrow & & \uparrow & & \\
0 & \longrightarrow & M^{i-1} & \longrightarrow & L^i \oplus M^{i-1} & \longrightarrow & L^i & \longrightarrow & 0 \\
 & & \uparrow & & \uparrow & & \uparrow & & \\
 & & \vdots & & \vdots & & \vdots & &
\end{array}$$

with a class $\mathrm{H}^{i-1}(L[1]^\bullet) = \mathrm{H}^i(L^\bullet)$ represented by $\ell \in L^i$ (such that $d^i_{L^\bullet}(\ell) = 0$) we find

$$\begin{array}{ccc}
\alpha^i(\ell) & \longmapsto & (0, \alpha^i(\ell)) \\
 & & \uparrow \\
 & & (\ell, 0) \longmapsto \ell
\end{array}$$

which verifies the claim. □

Corollary 7.1.1. *Let $\alpha^\bullet : L^\bullet \to M^\bullet$ be a morphism of complexes. Then the induced morphism $\mathrm{H}^\bullet(L^\bullet) \to \mathrm{H}^\bullet(M^\bullet)$ is an isomorphism if and only if the mapping cone $\mathrm{Cone}(\alpha)^\bullet$ is an exact complex.*

Proof. Using Proposition 7.1.1, consider the following segment of the long exact sequence

$$\begin{array}{l}
\cdots \longrightarrow \mathrm{H}^{i-1}(\mathrm{Cone}(\alpha)^\bullet) \longrightarrow \mathrm{H}^i(L^\bullet) \xrightarrow{\alpha^*} \\
\quad \mathrm{H}^i(M^\bullet) \longrightarrow \mathrm{H}^i(\mathrm{Cone}(\alpha)^\bullet) \longrightarrow \mathrm{H}^{i+1}(L^\bullet) \xrightarrow{\alpha^*} \\
\quad \mathrm{H}^{i+1}(M^\bullet) \longrightarrow \mathrm{H}^{i+1}(\mathrm{Cone}(\alpha)^\bullet) \longrightarrow \cdots
\end{array}$$

$\Longrightarrow$ Suppose α^* is an isomorphism. Since $\mathsf{Ch}(\mathsf{A})$ is abelian by Lemma 5.2.2 α^* is both a monomorphism and an epimorphism. Consider

$$\cdots \xrightarrow{\alpha^*} \mathrm{H}^i(M^\bullet) \xrightarrow{f} \mathrm{H}^i(\mathrm{Cone}(\alpha)^\bullet) \xrightarrow{g} \mathrm{H}^{i+1}(L^\bullet) \xrightarrow{\alpha^*} \cdots$$

Using that the first α^* is a surjective and exactness we have that

$$\ker f = \operatorname{im} \alpha^* = \mathrm{H}^i(M^\bullet)$$

i.e: f is the zero map.

Using that the last α^* is injective and exactness we have that

$$\ker g = \ker \alpha^* = \{0\}$$

i.e: g is the zero map.

Therefore we have that $\mathrm{H}^i \operatorname{Cone}(\alpha)^\bullet = 0$, meaning that the complex is exact.

$\Longleftarrow$ Suppose $\operatorname{Cone}(\alpha)^\bullet$ is an exact complex. Therefore since its cohomology vanishes at every index the sequence becomes

$$\cdots \longrightarrow 0 \longrightarrow \mathrm{H}^i(L^\bullet) \xrightarrow{\alpha^*} \mathrm{H}^i(M^\bullet) \longrightarrow 0 \longrightarrow \cdots$$

Therefore by exactness we have that α^* is an isomorphism.

□

7.2 Quasi-isomorphisms

Definition 7.2.1 (Quasi-isomorphism)**.** A morphism of complexes $\alpha^\bullet$ is a **quasi-isomorphism** (abbreviated q.i.s) if it induces an isomorphism in cohomology. ⌟

Remark 7.2.1. By Corollary 7.1.1, $\alpha^\bullet$ is a quasi-isomorphism if and only it its mapping cone $\operatorname{Cone}(\alpha)^\bullet$ is an exact complex. ▽

Example 7.2.1 (Quasi-isomorphism are not isomorphism)**.** In $\mathsf{Ch}(\mathsf{Ab})$, consider the morphism of complexes

$$\begin{array}{ccccccccccc}
\cdots & \longrightarrow & 0 & \longrightarrow & \mathbb{Z} & \xrightarrow{\cdot 2} & \mathbb{Z} & \longrightarrow & 0 & \longrightarrow & \cdots \\
 & & \downarrow & & \downarrow & & \downarrow{\scriptstyle \pi} & & \downarrow & & \\
\cdots & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \mathbb{Z}/_{2\mathbb{Z}} & \longrightarrow & 0 & \longrightarrow & \cdots
\end{array}$$

where π is the obvious projection is a quasi-isomorphism. However, it is not an isomorphism because there are no (non-trivial) group homomorphisms $\mathbb{Z}/_{2\mathbb{Z}} \to \mathbb{Z}$. ◊

Lemma 7.2.1. *Let* A *be an additive category and let* $F : \mathsf{Ch}(\mathsf{A}) \to \mathsf{D}$ *be an additive functor such that, for every quasi-isomorphism* $\rho^\bullet$, $F(\rho^\bullet)$ *is an isomorphism. Let* $M^\bullet$ *be an exact complex in* $\mathsf{Ch}(\mathsf{A})$.

- *The complex* $F(M^\bullet)$ *is a zero object in* D.
- *Let* $\alpha^\bullet : L^\bullet \to N^\bullet$ *be a morphism of complexes in* $\mathsf{Ch}(\mathsf{A})$ *that factors through an exact complex*

$$\begin{array}{ccc}
L^\bullet & \xrightarrow{\alpha^\bullet} & N^\bullet \\
 & \searrow \quad \nearrow & \\
 & M^\bullet &
\end{array}$$

  Then $F(\alpha^\bullet)$ *is the zero morphism.*

Proof. Since $M^\bullet$ is exact, the zero morphism $M^\bullet \to M^\bullet$ is a quasi-isomorphism. Therefore we have

$$F(M^\bullet) \xrightarrow{F(0)} F(M^\bullet) \xrightarrow{F(0)^{-1}} F(M^\bullet), \qquad \mathbb{1}_{F(M^\bullet)}$$

Since F is additive $F(0) = 0$, making $\mathbb{1}_{F(M^\bullet)} = 0$.

For the other part, applying F to the diagram we get

$$\begin{array}{ccc} F(L^\bullet) & \xrightarrow{F(\alpha^\bullet)} & F(N^\bullet) \\ & \searrow \quad \nearrow & \\ & F(M^\bullet) & \end{array}$$

Therefore since $F(M^\bullet)$ is a zero object in D we have that $F(\alpha^\bullet) = 0$. □

7.3 Homotopy

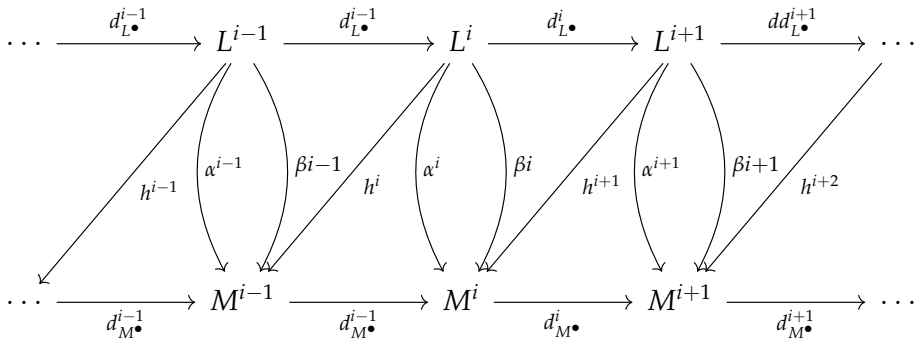
Definition 7.3.1. A **homotopy** h between two morphisms of complexes $\alpha^\bullet, \beta^\bullet : L^\bullet \to M^\bullet$ is a collection of morphisms

$$h^i : L^i \to M^{i-1}$$

such that

$$\forall i \in \mathbb{Z} \quad \beta^i - \alpha^i = d^{i-1}_{M^\bullet} \circ h^i + h^{i+1} \circ d^i_{L^\bullet}$$

Pictorially, we have the following **non-commutative** diagram



If there is an homotopy between $\alpha^\bullet$ and $\beta^\bullet$ we say that $\alpha^\bullet$ is **homotopic** to $\beta^\bullet$ and denote it by $\alpha^\bullet \sim \beta^\bullet$. ⌟

Definition 7.3.2. A morphism $\alpha^\bullet : L^\bullet \to M^\bullet$ is a **homotopy equivalence** if there exists a morphism $\beta^\bullet : M^\bullet \to L^\bullet$ such that $\alpha^\bullet \circ \beta^\bullet \sim \mathbb{1}_{M^\bullet}$ and $\beta^\bullet \circ \alpha^\bullet \sim \mathbb{1}_{N^\bullet}$. In this case, the complexes $L^\bullet$ and $M^\bullet$ are said to be **homotopy equivalent**. ⌟

Proposition 7.3.1. *If $\alpha^\bullet, \beta^\bullet : L^\bullet \to M^\bullet$ are homotopic morphisms of complexes then $\alpha^\bullet, \beta^\bullet$ induce the same morphisms on cohomology* $\mathrm{H}^\bullet(L^\bullet) \to \mathrm{H}^\bullet(M^\bullet)$.

Proof. Let $\overline{\ell} \in \mathrm{H}^i(L^\bullet)$. We have that $\overline{\ell}$ is represented by an element $\ell \in \ker d^i_{L^\bullet}$, and its images in $\mathrm{H}^i(M^\bullet)$ under α^*, β^* are represented by $\alpha^i(\ell), \beta^i(\ell)$.

Since $\alpha^\bullet, \beta^\bullet$ are homotopic there exist morphisms $h^i : L^i \to M^{i-1}$ such that

$$\beta^i(\ell) - \alpha^i(\ell) = d^{i-1}_{M^\bullet}(h^i(\ell)) + h^{i+1}(\cancelto{0}{d^i_{L^\bullet}(\ell)})$$

Therefore

$$\beta^i(\ell) - \alpha^i(\ell) \in \operatorname{im} d^i_{M^\bullet}$$

making $\beta^i(\ell) - \alpha^i(\ell)$ vanish in $\mathrm{H}^i(M^\bullet)$ as needed. □

Remark 7.3.1. If $\alpha^\bullet$ is homotopic to the identity by Corollary 7.1.1 the mapping cone of $\alpha^\bullet$ must be exact. ▽

Corollary 7.3.1. *Homotopy equivalent complexes have isomorphic cohomology.*

Proof. If $\alpha^\bullet : L^\bullet \to M^\bullet$ and $\beta^\bullet : M^\bullet \to L^\bullet$ are morphisms of complexes such that $\beta^\bullet \circ \alpha^\bullet$ and $\alpha^\bullet \circ \beta^\bullet$ are homotopic to the identity then by Proposition 7.3.1 they induce inverse morphisms in cohomology. □

Lemma 7.3.1. *Let $F : \mathsf{A} \to \mathsf{B}$ be an additive functor.*

- *If $\alpha^\bullet \sim \beta^\bullet$ are homotopic in $\mathsf{Ch}(\mathsf{A})$ then $\mathsf{Ch}(F)(\alpha^\bullet) \sim \mathsf{Ch}(F)(\beta^\bullet)$ in $\mathsf{Ch}(\mathsf{B})$.*
- *If $L^\bullet, M^\bullet$ are homotopy equivalent complexes in $\mathsf{Ch}(\mathsf{A})$ then $\mathsf{Ch}(F)(L^\bullet)$ and $\mathsf{Ch}(F)(M^\bullet)$ are homotopy equivalent complexes in $\mathsf{Ch}(\mathsf{B})$.*

Proof. Note that the second assertion follows directly from the first.

To see the first, if h is a homotopy between $\alpha^\bullet, \beta^\bullet$ then

$$\forall i \in \mathbb{Z} \quad \beta^i - \alpha^i = d^{i-1}_{M^\bullet} \circ h^i + h^{i+1} \circ d^i_{L^\bullet}$$

Since F is additive

$$\forall i \in \mathbb{Z} \quad F(\beta^i) - F(\alpha^i) = F(d^{i-1}_{M^\bullet}) \circ F(h^i) + F(h^{i+1}) \circ F(d^i_{L^\bullet})$$

showing that the collection $\{F(h^i)\}_{i\in\mathbb{Z}}$ produces a homotopy between $F(\alpha^\bullet)$ and $F(\beta^\bullet)$. □

The following is direct from the previous statements.

Theorem 7.3.1. *Let $F : \mathsf{A} \to \mathsf{B}$ be an additive between two abelian categories A, B. If $L^\bullet, M^\bullet$ are homotopy equivalent complexes in $\mathsf{Ch}(\mathsf{A})$ then the cohomology complexes*

$$\mathrm{H}^\bullet(\mathsf{Ch}(F)(L^\bullet)) \cong \mathrm{H}^\bullet(\mathsf{Ch}(F)(M^\bullet))$$

are isomorphic.

Part III

Sheaves

Chapter 8

Presheaves and sheaves

8.1 Presheaves

We begin by giving a general definition of a presheaf.

Definition 8.1.1 (Presheaf)**.** Let X be a topological space and let $\mathsf{Op}(X)$ be its `ex:open-set-category`. Let C be a Locally small category. The Locally small category of C-valued **presheaves** is the `ex:functor-category`

$$\mathsf{PSh}_X(\mathsf{C}) := [\mathsf{Op}(X)^{\mathrm{op}}, \mathsf{C}]$$

When the topological space is clear we will write $\mathsf{PSh}(\mathsf{C})$.

The elements of this category are Functors $\mathcal{F} : \mathsf{Op}(X)^{\mathrm{op}} \to \mathsf{C}$ and morphisms are **??**s $\phi : \mathcal{F} \to \mathcal{G}$. ⌟

Let's rewrite Definition 8.1.1 less abstractly: a C-valued Presheaf $\mathcal{F}$ is the collection of data

- For every open set $U \subseteq X$ an object $\mathcal{F}(U)$ of C called *sections of $\mathcal{F}$ over U*.
- For every inclussion of open sets $V \subseteq U$ a morphism $\mathrm{res}_{U,V} : \mathcal{F}(U) \to \mathcal{F}(V)$ in C called *restriction* such that
  - For every open set U, $\mathrm{res}_{U,U} = \mathbb{1}_U$.
  - For every inclussion of open sets $W \subseteq V \subseteq U$, $\mathrm{res}_{U,V} \circ \mathrm{res}_{V,W} = \mathrm{res}_{U,W}$.

A *morphism of Presheaves* $\phi : \mathcal{F} \to \mathcal{G}$ is a collection of morphisms $\{\phi(U) : \mathcal{F}(U) \to \mathcal{G}(U)\}_{U \in \mathsf{Op}(X)}$ in C making the following diagram commute:

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\phi(U)} & \mathcal{G}(U) \\ \downarrow{\scriptstyle \mathrm{res}_{U,V}} & & \downarrow{\scriptstyle \mathrm{res}_{U,V}} \\ \mathcal{F}(V) & \xrightarrow{\phi(V)} & \mathcal{G}(V) \end{array}$$

We are going to study now **??**s and **??**s in $\mathsf{PSh}(\mathsf{C})$.

Note that, since $\mathsf{PSh}(\mathsf{C})$ is a `def:functor-category` then if C has **??**s or **??**s of a certain shape then $\mathsf{PSh}(\mathsf{C})$ also has them and they are computed pointwise.

This means that **??**s and **??**s in $\mathsf{PSh}(\mathsf{C})$ can be computed *open set by open set*.

Proposition 8.1.1. *Let X be a topological space. Let $\mathcal{I} \to \mathsf{PSh}_X(\mathsf{C}), i \mapsto \mathcal{F}_i$ be a `def:diagram`. If C has **??**s and/or **??**s with shape $\mathcal{I}$ then $\lim_i \mathcal{F}_i$ and/or $\mathrm{colim}_i \mathcal{F}_i$ exist in $\mathsf{PSh}_X(\mathsf{C})$. Moreover*

for every open set $U \subseteq X$

$$\left(\lim_i \mathcal{F}_i\right)(U) = \lim_i \mathcal{F}_i(U) \qquad \left(\operatorname*{colim}_i \mathcal{F}_i\right)(U) = \operatorname*{colim}_i \mathcal{F}_i(U)$$

Write a section on functor categories (or quote it from somewhere) to justify this.

In particular if C is an Abelian category then Kernels and Cokernels exist and are computed *open set by open set*, making $\mathsf{PSh}(\mathsf{C})$ an Abelian category.

Remark 8.1.1. From this point onwards, we will suppose C has all **??**s and **??**s unless stated otherwise. ▽

8.1.1 Presheaves and continuous maps

Definition 8.1.2 (Direct image Presheaf)**.** Let $f : X \to Y$ be a continuous map. Let $\mathcal{F}$ be a C-valued Presheaf over X. The **direct image Presheaf** of $\mathcal{F}$ via f is the Presheaf $f_*\mathcal{F}$ defined as, for every open subset $V \subseteq Y$

$$f_*\mathcal{F}(V) = \mathcal{F}(f^{-1}(U))$$

Given $W \subseteq V \subseteq Y$ open subsets, the restriction map comes from the commutativity of the diagram

$$\begin{array}{ccc} f_*\mathcal{F}(V) & = & \mathcal{F}(f^{-1}V) \\ \vdots & & \downarrow \\ f_*\mathcal{F}(V) & = & \mathcal{F}(f^{-1}V) \end{array}$$

This construction defines a Functor $f_* : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{PSh}_Y(\mathsf{C})$. ⌟

Show that the category of presheaves over a one point space *is equal* to the evaluation category.

Example 8.1.1 (Global section)**.** Let X be a topological space. Consider the one-point topological space $\{*\}$. Let $\mathcal{F}$ be a C-valued presheaf over X. The **global section** of $\mathcal{F}$ over

This defines a Functor $\Gamma(X, -) : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{C}$ ◊

Example 8.1.2 (Skyscrapper Presheaf)**.** Let X be a topological space and $p \in X$. Suppose C has a final object $* \in \mathsf{C}$. Let S be an object in C. The **skyscrapper Presheaf** supported at S $\iota_{p,*}\underline{S}_{\text{pre}}$ is the Direct image Presheaf of the Constant Presheaf $\underline{S}_{\text{pre}}$ via the inclussion $\iota : \{p\} \hookrightarrow X$:

$$(\iota_{p,*}\underline{S}_{\text{pre}})(U) = \begin{cases} S & p \in U \\ * & p \notin U \end{cases}$$

◊

Using this we can construct a *dual* Presheaf to Definition 8.1.2.

Definition 8.1.3 (Inverse image Presheaf)**.** Let $f : X \to Y$ be a continuous map. Let $\mathcal{G}$ be a C-valued Presheaf over Y. The **inverse image Presheaf** of $\mathcal{G}$ via f is the Presheaf $f^{-1}\mathcal{G}$ defined as, for every open subset $U \subseteq X$

$$f^{-1}\mathcal{G}(U) = \operatorname*{colim}_{V \supseteq f(U)} \mathcal{G}(V)$$

where the def:categorical-colimit is taken over all the open neighbourhoods V of $f(U)$.

This defines a Functor $f^{-1} : \mathsf{PSh}_Y(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$. ⌟

Remark 8.1.2. Note that the set of open neighbourhoods of $f(U)$ is a partially ordered set via reverse inclussion. Therefore if C is a concrete category we can describe $f^{-1}\mathcal{G}(U)$ as $\{(s, V) \mid f(U) \subseteq V \in \mathsf{Op}(Y), s \in \mathcal{G}(V)\}$ modulo the relation $(s, V) \sim (s', V')$ if and only if there exists $W \subseteq V \cap V'$ open with $f(U) \subseteq W$ such that $s|_W = s'|_W$. ▽

Example 8.1.3 (Constant Presheaf)**.** Let X be a topological space. Let S be an object of C. The **constant Presheaf** associated to S $\underline{S}_{\text{pre}}$ is the Inverse image Presheaf of S via the map $\{*\} \to X$. ◊

Definition 8.1.4 (Stalk)**.** Let X be a topological space and $p \in X$. Let $\mathcal{F}$ be a C-valued Presheaf over X. The **stalk** of $\mathcal{F}$ at p $\mathcal{F}_p$ is the Inverse image Presheaf of $\mathcal{F}$ via $X \to \{p\}$:

$$\mathcal{F}_p = \operatorname*{colim}_{U \ni p} \mathcal{F}(U)$$

⌟

Proposition 8.1.2. *Let $f : X \to Y$ be a continuous map. We have an def:adjunction between the Direct image Presheaf functor and the Inverse image Presheaf $f^{-1} \dashv f_*$.*

Proof. We define the def:categorical-unit $\eta : 1 \implies f_* f^{-1}$ as the map with components given by the canonical insertion maps to the colimit

$$\mathcal{F}(W) \to \operatorname*{colim}_{U \supseteq (f \circ f^{-1})(W)} \mathcal{F}(U) = f^{-1}\mathcal{F}(f^{-1}W) = f_* f^{-1}\mathcal{F}(W)$$

Note that this maps exist because $(f \circ f^{-1})(W) \subseteq W$.

On the other hand, note that if $f(V) \subseteq U$ then $V \subseteq (f^{-1} \circ f)(V) \subseteq f^{-1}(U)$, i.e: we have a restricion map $\mathcal{G}(f^{-1}(U)) \to \mathcal{G}(V)$. Therefore by the universal property of the colimit we have a unique map

$$f^{-1} f_* \mathcal{G}(V) = \operatorname*{colim}_{U \supseteq f(V)} \mathcal{G}(f^{-1}(U)) \to \mathcal{G}(V)$$

Using this maps we obtain a def:categorical-counit $\varepsilon : f^{-1} f_* \implies 1$.

Since by construction the compositions $\eta \circ \varepsilon$ and $\varepsilon \circ \eta$ are the corresponding identity morphisms, we have proven the adjunction. □

Two important particular cases of the adjunction Proposition 8.1.2 are

Corollary 8.1.1. *Let X be a topological space and $p \in X$. Let $\mathcal{F}$ be a C-valued Presheaves over X and let A be an object in C. Then*

$$\mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\underline{A}_{\text{pre}}, \mathcal{F}) \simeq \mathrm{Hom}_{\mathsf{C}}(A, \Gamma(X, \mathcal{F}))$$

Corollary 8.1.2. *Let X be a topological space and $p \in X$. Let $\mathcal{F}$ be a C-valued Presheaves over X and let A be an object in C. Then*

$$\mathrm{Hom}_{\mathsf{C}}(\mathcal{F}_p, A) \simeq \mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\mathcal{F}, \iota_{p,*}\underline{A}_{\text{pre}})$$

8.2 Sheaves

We now turn our attention to a particular class of Presheaves, following [**stacks-project**].

Let's start by considering the case $\mathsf{C} = \mathsf{Set}$:

Definition 8.2.1 (Sheaf of sets)**.** Let X be a topological space. A Presheaf of sets $\mathcal{F}$ is a **sheaf** if for every open set U and open covering $\{U_i\}_{i\in\mathcal{I}}$ of U the diagram

$$\mathcal{F}(U) \to \prod_{i\in\mathcal{I}} \mathcal{F}(U_i) \rightrightarrows \prod_{(i,j)\in\mathcal{I}\times\mathcal{I}} \mathcal{F}(U_i \cap U_j) \tag{8.2.1}$$

is an `def:equalizer`.

We denote by $\mathsf{Sh}_X(\mathsf{Set})$ (or just Sh if the context is clear) the `def:full-subcategory` of the category of presheaves of sets whose objects are sheaves of sets.

Less abstractly this means that $\mathcal{F}$ verifies:

SH.1) **Identity**: Given two sections $s, t \in \mathscr{F}(U)$ such that for every $i \in \mathcal{I}$ $\quad s|_{U_i} = t|_{U_i}$ we have that $s = t$.

SH.2) **Gluability**: Given a collection of sections $\{s_i \mid s_i \in \mathscr{F}(U_i)\}_{i\in\mathcal{I}}$ that agree on pairwise intersections ($\forall i, j \in \mathcal{I}\ \ s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j}$) there exists a section $s \in \mathscr{F}(U)$ such that $\forall i \in \mathcal{I}\ \ s|_{U_i} = s_i$.

⌟

Definition 8.2.2 (Locally small category of Sheaves of sets)**.** The Locally small category $\mathsf{Sh}_X(\mathsf{Set})$ of Sheaves of sets is the `def:full-subcategory` of the category of Presheaves $\mathsf{PSh}_X(\mathsf{Set})$ whose objects are Sheaves of sets. ⌟

Let's try to make Definition 8.2.1 more general by considering another evaluation category C.

Firstly, note that if **??**s indexed by $\mathcal{I}$ and $\mathcal{I}\times\mathcal{I}$ for any indexing set $\mathcal{I}$ exist in C then the diagram (8.2.1) makes sense. Secondly, we have that (8.2.1) is an `def:equalizer` if and only if for every object $C \in \mathsf{C}$ the diagram in Set

$$\mathrm{Hom}_{\mathsf{C}}(C, \mathcal{F}(U)) \to \prod_{i\in\mathcal{I}} \mathrm{Hom}_{\mathsf{C}}(C, \mathcal{F}(U_i)) \rightrightarrows \prod_{(i,j)\in\mathcal{I}\times\mathcal{I}} \mathrm{Hom}_{\mathsf{C}}(C, \mathcal{F}(U_i \cap U_j))$$

is an `def:equalizer`.

Therefore we get

Definition 8.2.3 (Sheaf)**.** Let X be a topological space. Let C be an arbitrary category. A C-valued Presheaf $\mathcal{F}$ is a **sheaf** if for all objects $C \in \mathsf{C}$ the Presheaf of sets $\mathcal{F}_C$ defined as $\mathcal{F}_C(U) = \mathrm{Hom}_{\mathsf{C}}(C, \mathcal{F}(U))$ is a Sheaf of sets.

We denote by $\mathsf{Sh}_X(\mathsf{C})$ the `def:full-subcategory` of the category of C-valued presheaves of sets whose objects are C-valued sheaves. ⌟

Notation. From now on, to distinguish a Presheaf from a Sheaf we will denote the former by $\mathcal{F}, \mathcal{G}, \mathcal{H}, \ldots$ and the latter by $\mathscr{F}, \mathscr{G}, \mathscr{H}, \ldots$

By Definition 8.2.3 we have an inclussion functor $\iota : \mathsf{Sh}(\mathsf{C}) \hookrightarrow \mathsf{PSh}(\mathsf{C})$.

For *sufficiently nice* Locally small categorys, this Functor admits a left-adjoint $\mathsf{PSh}(\mathsf{C}) \to \mathsf{Sh}(\mathsf{C})$. Let's study first the case $\mathsf{C} = \mathsf{Set}$.

How general do we want to go?

Theorem 8.2.1 (Sheafification). *There exists a* ***sheafification*** *Functor* $(-)^{\#} : \mathsf{PSh}(\mathsf{Set}) \to \mathsf{Sh}(\mathsf{Set})$ *that is left-adjoint to the inclussion Functor* $\iota : \mathsf{Sh}(\mathsf{Set}) \to \mathsf{PSh}(\mathsf{Set})$.

Proof. Let $\mathcal{F}$ be a Presheaf of sets over X. For every open subset $U \subseteq X$ we define

$$\mathcal{F}^{\#}(U) = \left\{ f_p \in \prod_{p \in U} \mathcal{F}_p \text{ verifying } (*) \right\}$$

where $(*)$ is the property *"for all $p \in U$ there exists an open neighbourhood $V \subseteq U$ of p and a section $s \in \mathcal{F}(V)$ such that, for every $q \in V$, $s_q = f_q$"*.

Taking $\prod_{p \in U} \mathcal{F}_p \to \prod_{q \in V} \mathcal{F}_q$ as the restriction maps we see that $\mathcal{F}^{\#}$ is a Presheaf. Let's see now that $\mathcal{F}^{\#}$ is a Sheaf. To do this, consider an open subset $U \subseteq X$ with open cover $\{U_i\}_{i \in \mathcal{I}}$.

SH.1 Let $s, t \in \mathcal{F}^{\#}(U)$ be two sections such that, for all $i \in \mathcal{I}$ $s|_{U_i} = t|_{U_i}$.

Then since for every $p \in U$ there exists $i \in \mathcal{I}$ such that $p \in U_i$ we have that $s_p = (s|_{U_i})_p = (t|_{u_i})_p = t_p$. Therefore $s = t$.

SH.2 Let $\{s_i \in \mathcal{F}^{\#}(U_i)\}_{i \in \mathcal{I}}$ be a collection of sections such that $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all $i, j \in \mathcal{I}$.

Consider $s \in \prod_{p \in U} \mathcal{F}_p$ defined as $s_p = (s_i)_p$ if $p \in U_i$. Note that this is well-defined by property $(*)$. To see that s verifies $(*)$ note that for every $p \in U_i$ there exists an open neighbourhood $V \subseteq U$ of p and $t \in \mathcal{F}(V)$ such that, for all $q \in V$, $t_q = (s_i)|_q = s_q$.

Let $V \subseteq U \subseteq X$ be two open sets. Consider the commutative diagram

$$\begin{array}{ccccc} \mathcal{F}(U) & \longrightarrow & \mathcal{F}^{\#}(U) & \longrightarrow & \prod_{p \in U} \mathcal{F}_p \\ \downarrow & & \downarrow & & \downarrow \\ \mathcal{F}(V) & \longrightarrow & \mathcal{F}^{\#}(V) & \longrightarrow & \prod_{q \in V} \mathcal{F}_q \end{array}$$

where the vertical arrows are the corresponding restriction maps. By chasing this diagram we get a map $\varphi : \mathcal{F} \to \mathcal{F}^{\#}$, $\varphi_U(s) = (s_p)_{p \in U}$.

It only remains to show that any morphism $\psi : \mathcal{F} \to \mathcal{G}$, where $\mathcal{G}$ is a Sheaf of sets, factors uniquely as $\mathcal{F} \xrightarrow{\varphi} \mathcal{F}^{\#} \to \mathcal{G}$, i.e: we have to construct a morphism $\theta : \mathcal{F}^{\#} \to \mathcal{G}$ such that $\psi = \theta \circ \varphi$.

Let $U \subseteq X$ be an open subset and $s \in \mathcal{F}^{\#}(U)$. For each $p \in U$ consider an open neighbourhood $V^{(p)} \subseteq U$ of p and $t^{(p)} \in \mathcal{F}(V^{(p)})$ such that $t^{(p)}_q = s_q$. Consider now $\psi_{V^{(p)}}(t^{(p)}) \in \mathcal{G}(V^{(p)})$. Note that $\{\psi_{V^{(p)}}(t^{(p)})\}_{p \in U}$ is a compatible family of section in $\mathcal{G}$, therefore by the Sheaf axioms there exists a unique section $t \in \mathcal{G}(U)$ such that $t|_{V^{(p)}} = \psi_{V^{(p)}}(s^{(p)})$. Therefore we can define $\theta_U(s) = t$.

By construction $\psi = \theta \circ \varphi$ and θ commutes with the restriction maps, making it a morphism of sheaves.

To check uniqueness, let $\theta' : \mathcal{F}^{\#} \to \mathcal{G}$ be another morphism such that $\psi = \theta' \circ \varphi$. Consider $s \in \mathcal{F}^{\#}(U)$. For each $p \in U$ choose $V^{(p)} \subseteq U$ and $s^{(p)} \in \mathcal{F}(V^{(p)})$ such that $s|_{V^{(p)}} = \varphi_{V^{(p)}}(s^{(p)})$. Then

$$\theta'(s)|_{V^{(p)}} = \theta'(s|_{V^{(p)}}) = \theta'(\varphi_{V^{(p)}}(s^{(p)})) = \psi_{V^{(p)}}(s^{(p)})$$

On the other hand $\theta(s)|_{V^{(p)}} = \psi_{V^{(p)}}(s^{(p)})$ by definition. Therefore since $\mathscr{G}$ is a Sheaf by SH.1 we have $\theta'(f) = \theta(f)$, i.e: $\theta' = \theta$. □

We can extend the construction outlined in Theorem 8.2.1 to **??**s.

Lemma 8.2.1. *Let (C, F) be an* `def:algebraic-category`*. Let $\mathcal{F}$ be a C-valued Presheaf. Then there exists a C-valued Sheaf $\mathcal{F}^{\#}$ and a morphism of presheaves $\mathcal{F} \to \mathcal{F}^{\#}$ satisfying*

1. *The map $\mathcal{F} \to \mathcal{F}^{\#}$ identifies the underlying Sheaf of sets of $\mathcal{F}^{\#}$ with the sheafification of the underlying Presheaf of sets of $\mathcal{F}$.*
2. *For any morphism of presheaves $\mathcal{F} \to \mathscr{G}$ where $\mathscr{G}$ is a C-valued Sheaf there exists a unique factorization $\mathcal{F} \to \mathcal{F}^{\#} \to \mathscr{G}$.*

Remark 8.2.1. We will assume that C is an `def:algebraic-category` in what follows. ▽

Therefore we can assign a Sheaf to any Presheaf using this construction.

Definition 8.2.4 (Sheafification)**.** Let $\mathcal{F}$ be a C-valued Presheaf. We call the Sheaf $\mathcal{F}^{\#}$ obtained by Theorem 8.2.1 the **sheafification** of $\mathcal{F}$. ⌟

Using this adjunction we can *adapt* Definition 8.1.2 and Definition 8.1.3 to the case of Sheaves

Definition 8.2.5 (Direct image Sheaf)**.** Let $f : X \to Y$ be a continuous map. Let $\mathscr{F}$ be a C-valued Sheaf over X. The **direct image Sheaf** of $\mathscr{F}$ via f is the Sheaf $f_*\mathscr{F}$ defined as, for every open subset $V \subseteq Y$

$$f_*\mathscr{F}(V) = \mathscr{F}(f^{-1}(U))$$

This construction defines a Functor $f_* : \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{Sh}_Y(\mathsf{C})$. ⌟

This means that Example 8.1.1 and Example 8.1.2 translate directly to Sheaves.

Definition 8.2.6 (Global section Functor)**.** We denote by $\Gamma(X, -) : \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{C}$ the **global section functor** described in Example 8.1.1, but now with initial category $\mathsf{Sh}_X(\mathsf{C})$. ⌟

Definition 8.2.7 (Skyscrapper Sheaf)**.** Let X be a topological space and $p \in X$. Let S be an object of C. The **skyscrapper Sheaf** supported at p $\iota_{*,p}\underline{S}$ is the Direct image Sheaf of the Constant Sheaf $\underline{S}$ via the map $\{p\} \to X$. ⌟

Rewrite this definitions

Explain why we need to sheafificate

Definition 8.2.8 (Inverse image Sheaf)**.** Let $f : X \to Y$ be a continuous map. Let $\mathscr{G}$ be a C-valued Sheaf over Y. The **inverse image Sheaf** of $\mathscr{G}$ via f is the Sheaf $f^{-1}\mathscr{G}$ defined as, for every open subset $U \subseteq X$

$$f^{-1}\mathscr{G} = (f^{-1}\mathscr{G})^{\#}$$

where the RHS of the equation is the Sheafification of the Inverse image Presheaf of $\mathscr{G}$ via f.

This defines a Functor $f^{-1} : \mathsf{PSh}_Y(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$. ⌟

Definition 8.2.9 (Constant Sheaf)**.** Let S be an object of C. The **constant Sheaf** $\underline{S}$ associated to S is the Sheafification of the Constant Presheaf $\underline{S}_{\text{pre}}$, i.e: $\underline{S} = (\underline{S}_{\text{pre}})^{\#}$. ⌟

Definition 8.2.10 (Stalk)**.** Let X be a topological space and $p \in X$. Let $\mathscr{F}$ be a C-valued Sheaf over X. The **stalk** of $\mathscr{F}$ at p $\mathscr{F}_p$ is the Inverse image Sheaf of $\mathscr{F}$ via $X \to \{p\}$:

$$\mathscr{F}_p = \operatorname*{colim}_{U \ni p} \mathscr{F}(U)$$

⌟

Using the adjointness we have the analogous of Proposition 8.1.2

Proposition 8.2.1. *Let $f : X \to Y$ be a continuous map. We have an* `def:adjunction` *between the Direct image Sheaf functor and the Inverse image Sheaf $f^{-1} \dashv f_*$.*

Proof.

$$\begin{aligned}\operatorname{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(f^{-1}\mathscr{G}, \mathscr{F}) &= \operatorname{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(f^{-1}\mathscr{G}, \mathscr{F}) \\ &= \operatorname{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\mathscr{G}, f_*\mathscr{G}) \\ &= \operatorname{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(\mathscr{G}, f_*\mathscr{G})\end{aligned}$$

□

Therefore adjunctions Corollary 8.1.1 and Corollary 8.2.2 also happen when considering Sheaves.

Corollary 8.2.1. *Let X be a topological space and $p \in X$. Let $\mathscr{F}$ be a C-valued Sheaves over X and let A be an object in C. Then*

$$\operatorname{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(\underline{A}, \mathscr{F}) \simeq \operatorname{Hom}_{\mathsf{C}}(A, \Gamma(X, \mathscr{F}))$$

Corollary 8.2.2. *Let X be a topological space and $p \in X$. Let $\mathscr{F}$ be a C-valued Sheaves over X and let A be an object in C. Then*

$$\operatorname{Hom}_{\mathsf{C}}(\mathscr{F}_p, A) \simeq \operatorname{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(\mathscr{F}, \iota_{p,*}\underline{A})$$

For cohomology we will be interested in the case $\mathsf{C} = \mathsf{Ab}$.

Lemma 8.2.2. *Let $a : \mathsf{A} \to \mathsf{B}$ and $b : \mathsf{B} \to \mathsf{A}$ be two Functors. Suppose that*

1. *A and B are additive categories, a, b are additive functors and $b \dashv a$.*
2. *B is an Abelian category and b is left-exact.*
3. *$ba \cong \mathbb{1}_{\mathsf{A}}$.*

Then A is abelian.

Proof. Since B is an Abelian category all **??**s and **??**s exist en B.

Since b is left-adjoint it is right-exact and therefore, exact.

Let $\varphi : B_1 \to B_2$ be a morphism in B. If $K = \ker(\varphi)$ then K is the `def:equalizer` of the diagram

$$K \longrightarrow B_1 \underset{0}{\overset{\varphi}{\rightrightarrows}} B_2$$

therefore applying b we get

$$bK \longrightarrow bB_1 \underset{0}{\overset{b\varphi}{\rightrightarrows}} bB_2$$

i.e: bK is the Kernel of $b\varphi$.

Dually we get that $b\operatorname{coker}(B_1 \to B_2)$ is the Cokernel of $b\varphi$.

Therefore we have that every morphism $b\varphi$ in A admits a Kernel and a Cokernel. Moreover since $ba \cong \mathbb{1}_{\mathsf{A}}$ every morphism in A is of the form $b\varphi$, therefore all Kernels and Cokernels exist in A. Indeed for every $\psi : A_1 \to A_2$ we have that

$$\ker \psi = b\ker(a\psi) \qquad \operatorname{coker} \psi = b\operatorname{coker}(a\psi) \tag{8.2.2}$$

It only remains to show that $\operatorname{coim} \psi = \operatorname{im} \psi$. Since A has all **??**s and **??**s A has all finite **??**s and **??**s, making a left-exact. This means that a transforms Kernels into Kernels. Therefore

$$\begin{aligned}
\operatorname{coim} \psi &= \operatorname{coker}(\ker \psi \to A_1) && \text{(definition)} \\
&= b\operatorname{coker}(a(\ker \psi \to A_1)) && \text{(8.2.2)} \\
&= b\operatorname{coker}(\ker(a\psi) \to aA_1) && (a \text{ preserves kernels}) \\
&= b\operatorname{coim}(a\psi) && \text{(definition)} \\
&= b\operatorname{im}(a\psi) && (\mathsf{B} \text{ is abelian}) \\
&= b\ker(aA_2 \to \operatorname{coker}(a\psi)) && \text{(definition)} \\
&= \ker(baA_2 \to b\operatorname{coker}(a\psi)) && (b \text{ preserves kernels}) \\
&= \ker(A_2 \to b\operatorname{coker}(a\psi)) && ba \cong \mathbb{1}_{\mathsf{A}} \\
&= \ker(A_2 \to \operatorname{coker} \psi) && \text{(8.2.2)} \\
&= \operatorname{im} \psi && \text{(definition)}
\end{aligned}$$

□

Proposition 8.2.2. *Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism in $\mathsf{Sh}(\mathsf{Ab})$.*

1. *The Locally small category $\mathsf{Sh}_X(\mathsf{Ab})$ is an Abelian category.*
2. *The Kernel of φ is the same as the Kernel of φ in $\mathsf{PSh}(\mathsf{Ab})$.*
3. *φ is injective if and only if φ is injective as a map or presheaves if and only if φ is injective as a map of sheaves.*
4. *The Cokernel of φ is the Sheafification of the Cokernel of φ in $\mathsf{PSh}(\mathsf{Ab})$.*
5. *φ is surjective if and only if φ is surjective as a map of sheaves.*
6. *A complex $\mathscr{F} \to \mathscr{G} \to \mathscr{H}$ is exact at $\mathscr{G}$ if and only if for all open sets U and sections $s \in \mathscr{G}(U)$ mapping to 0 in $\mathscr{H}(U)$ there exists an open covering $\{U_i\}_{i\in\mathcal{I}}$ such that each $s|_{U_i}$ is the image of $\mathscr{F}(U_i) \to \mathscr{G}(U_i)$.*

Proof. Note that 2. $\implies$ 3., 4. $\implies$ 5 and 2. $\wedge$ 5. $\implies$ 6.

Therefore we only need to prove 1., 2. and 3., but this comes directly from Lemma 8.2.2 with $\mathsf{A} = \mathsf{Sh}_X(\mathsf{Ab})$, $\mathsf{B} = \mathsf{PSh}_X(\mathsf{Ab})$, $a : \mathsf{Sh}_X(\mathsf{Ab}) \hookrightarrow \mathsf{PSh}_X(\mathsf{Ab})$ the inclussion Functor and $b : \mathsf{PSh}_X(\mathsf{Ab}) \to \mathsf{Sh}_X(\mathsf{Ab})$ the Sheafification Functor. □

Chapter 9

Ringed spaces and $\mathscr{O}_X$-modules.

9.1 Ringed spaces

Definition 9.1.1 (Ringed space)**.** A **ringed space** is a tuple $(X, \mathscr{O}_X)$ where X is a topological space and $\mathscr{O}_X$ is a Sheaf of rings over X.

We call $\mathscr{O}_X$ the *structure sheaf* of the ringed space $(X, \mathscr{O}_X)$. Sections of $\mathscr{O}_X$ over an open subset $U \subseteq X$ are called *functions on* U and functions on X are called (global) functions. ⌟

Example 9.1.1. Let and X be a topological space. Then the $\mathbb{R}$-Algebra of continuous functions $\mathcal{C}^0_X(U) = \{f : U \to \mathbb{R} \mid f \text{ continuous}\}$ with the usual restriction maps forms a Sheaf of rings over X. Therefore $(X, \mathcal{C}^0_X)$ is a Ringed space. ◊

Example 9.1.2. If for every point $p \in X$ the Stalks $\mathscr{O}_{X,p}$ of the structure sheaf are Local rings then we call $(X, \mathscr{O}_X)$ a **locally ringed space**. ◊

Definition 9.1.2 (Morphism of Ringed spaces)**.** Let $(X, \mathscr{O}_X)$ and $(Y, \mathscr{O}_Y)$ be two Ringed spaces. A **morphism of ringed spaces** $(X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ is a pair $(f, f^\sharp)$ where $f : X \to Y$ is a continuous function (i.e: a morphism in Top) and $f^\sharp : \mathscr{O}_Y \to f_*\mathscr{O}_X$ is a morphism of sheaves of rings (i.e: a morphism in $\mathsf{Sh}_{\mathsf{Ring}}(X)$), where $f_*\mathscr{O}_X$ is the Direct image Sheaf. ⌟

Maybe it is better to define $f^\sharp$ as an f-map. **[stacks-project]**

Example 9.1.3. Let $f : X \to Y$ be a continuous function. Consider the Ringed spaces of continuous functions $(X, \mathcal{C}^0_X)$, $(Y, \mathcal{C}^0_Y)$.

Then the Morphism of Ringed spaces induced by f can be defined for every $V \subseteq Y$ open as $f^\sharp(V) : \mathcal{C}^0_Y(V) \to \mathcal{C}^0_X(f^{-1}(V))$, $f^\sharp(V)(h) = h \circ f|_{f^{-1}V}$. ◊

9.2 $\mathscr{O}_X$ modules

Definition 9.2.1 ($\mathscr{O}_X$-module)**.** Let $(X, \mathscr{O}_X)$ be a Ringed space. A **Sheaf of $\mathscr{O}_X$ modules** (or simply an $\mathscr{O}_X$-module) is a Sheaf $\mathscr{F}$ such that for every open set $U \subseteq X$ $\mathscr{F}(U)$ is an $\mathscr{O}_X(U)$-Module compatible with the restriction maps, i.e: for every $f \in \mathscr{O}_X(U)$ and $s \in \mathscr{F}(U)$: $(f \cdot s)|_V = f|_V \cdot s|_V$. ⌟

Example 9.2.1. ... ◊

Definition 9.2.2 (Morphism of $\mathscr{O}_X$-modules)**.** Let $\mathscr{F}, \mathscr{G}$ be two $\mathscr{O}_X$-modules. A morphism $\phi : \mathscr{F} \to \mathscr{G}$ is a sheaf morphism that commutes with the multiplication of the structure sheaf $\mathscr{O}_X$. ⌟

Example 9.2.2. ... ◊

Therefore for every Ringed space $(X, \mathscr{O}_X)$ the collection of $\mathscr{O}_X$-modules forms a Locally small category $\mathscr{O}_X$-Mod.

9.2.1 $\mathscr{O}_X$-Mod and sheaves of abelian groups.

Just as abelian groups are $\mathbb{Z}$-modules, Sheaves of abelian groups can be seen as $\underline{\mathbb{Z}}_X$-modules, where $\underline{\mathbb{Z}}_X$ is the `ex:constant-sheaf` with coefficients in the integers.

To show this, for every Sheaf of abelian groups $\mathscr{F}$ over X and every open set $U \subseteq X$ we have to define an action $\cdot : \underline{\mathbb{Z}}_X(U) \times \mathscr{F}(U) \to \mathscr{F}(U)$ compatible with the restriction maps.

Let $f \in \underline{\mathbb{Z}}_X(U)$ and $s \in \mathscr{F}(U)$. Since f is locally constant there exists an open cover $\{U_i\}_{i\in\mathcal{I}}$ of U such that for every $i \in \mathcal{I} \quad f|_{U_i} = n_i \in \mathbb{Z}$.

For every U_i consider $t_i = n_i \cdot s|_{U_i}$, where $n_i \cdot s|_{U_i}$ is shorthand for $s|_{U_i}$ added to itself n_i times. Note that this is well-defined because $\mathscr{F}(U_i)$ is an abelian group.

Let $U_i, U_j \in \{U_i\}_{i\in\mathcal{I}}$ such that $U_i \cap U_j = \varnothing$. For every $x \in U_i \cap U_j$ there exists an open neighbourhood $V \subseteq U_i \cap U_j$ of x were f is constant.

Therefore $n_i = n_j$ and $t_i|_{U_i\cap U_j} = t_j|_{U_i\cap U_j}$ and, applying axiom SH.2, we can define $f \cdot s = t$.

Checking the axioms in small enough open sets such that f is constant verifies that this action makes $\mathscr{F}$ an $\underline{\mathbb{Z}}_X$-module.

Do we have to actually write the check?

9.3 $\mathscr{O}_X$-Mod is abelian.

Let's check first that $\mathscr{O}_X$-Mod is a Additive category.

Lemma 9.3.1. *The category $\mathscr{O}_X$-Mod admits a zero object.*

Proof. Consider the $\mathscr{O}_X$ module that sends every open set to $\{0\}$. □

Lemma 9.3.2. *The category $\mathscr{O}_X$-Mod has all binary biproducts. We define their direct sum a $\mathscr{F} \oplus \mathscr{G} = \mathscr{F} \times \mathscr{G}$.*

Proof. Let $\mathscr{F}, \mathscr{G}$ be two $\mathscr{O}_X$-modules. □

Theorem 9.3.1. *Let $\varphi : \mathscr{F} \to \mathscr{G}$ a Morphism of $\mathscr{O}_X$-modules. Then the kernel of φ is the subsheaf of $\mathscr{F}$ with sections*

$$(\ker\varphi)(U) = \{s \in \mathscr{F}(U) \mid \varphi(s) = 0\}$$

Furthermore for every $x \in X \quad (\ker\varphi)_x = \ker\varphi_x$.

Proof. If $V \subseteq U$

$$\begin{array}{ccccccc} 0 & \longrightarrow & (\ker\varphi)(U) & \longrightarrow & \mathscr{F}(U) & \xrightarrow{\varphi(U)} & \mathscr{G}(U) \\ & & \vdots & & \downarrow{\scriptstyle \mathrm{res}_{U,V}} & & \downarrow{\scriptstyle \mathrm{res}_{U,V}} \\ 0 & \longrightarrow & (\ker\varphi)(V) & \longrightarrow & \mathscr{F}(V) & \xrightarrow{\varphi(V)} & \mathscr{G}(V) \end{array}$$

□

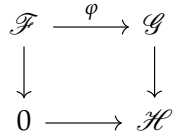
Theorem 9.3.2. *Let $\varphi : \mathscr{F} \to \mathscr{G}$ a Morphism of $\mathscr{O}_X$-modules. Then the cokernel of φ is the Sheafification of the quotient Presheaf*

$$(\operatorname{coker}_{\text{pre}} \varphi)(U) = {}^{\mathscr{G}(U)}\!/_{\varphi(\mathscr{F}(U))}$$

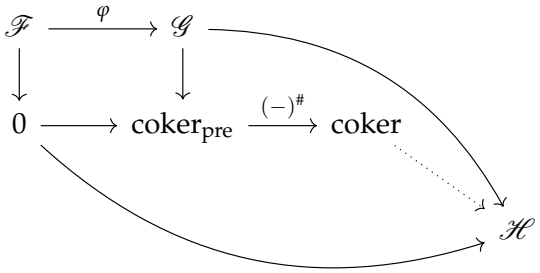
Furthermore for every $x \in X$ $\quad (\operatorname{coker} \varphi)_x = \operatorname{coker} \varphi_x$.

Prove that this is the cokernel in PSh.

Proof. Given a commutative diagram



We construct



□

Therefore we have proven that $\mathscr{O}_X$-Mod is an Abelian category and in particular by $\mathscr{O}_X$-Mod and sheaves of abelian groups $\mathsf{Sh}_{\mathsf{Ab}}(X)$ is an Abelian category.

9.4 Sequences in $\mathscr{O}_X$-Mod

Theorem 9.4.1. *The sequence of $\mathscr{O}_X$-modules $\mathscr{F} \to \mathscr{G} \to \mathscr{H}$ is exact at $\mathscr{G}$ if and only if for every $x \in X$ the sequence $\mathscr{F}_x \to \mathscr{G}_x \to \mathscr{H}_x$ is exact at $\mathscr{G}_x$.*

Probably not a *theorem* but a direct consequence of the commutatitivity of stalks and (co)kernels

Maybe talk about resolutions / types of resolutions here?

Appendix A

Algebra

Definition A.0.1 (Algebra)**.** ⌟

Definition A.0.2 (Local ring)**.** ⌟

Definition A.0.3 (Module)**.** ⌟

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